

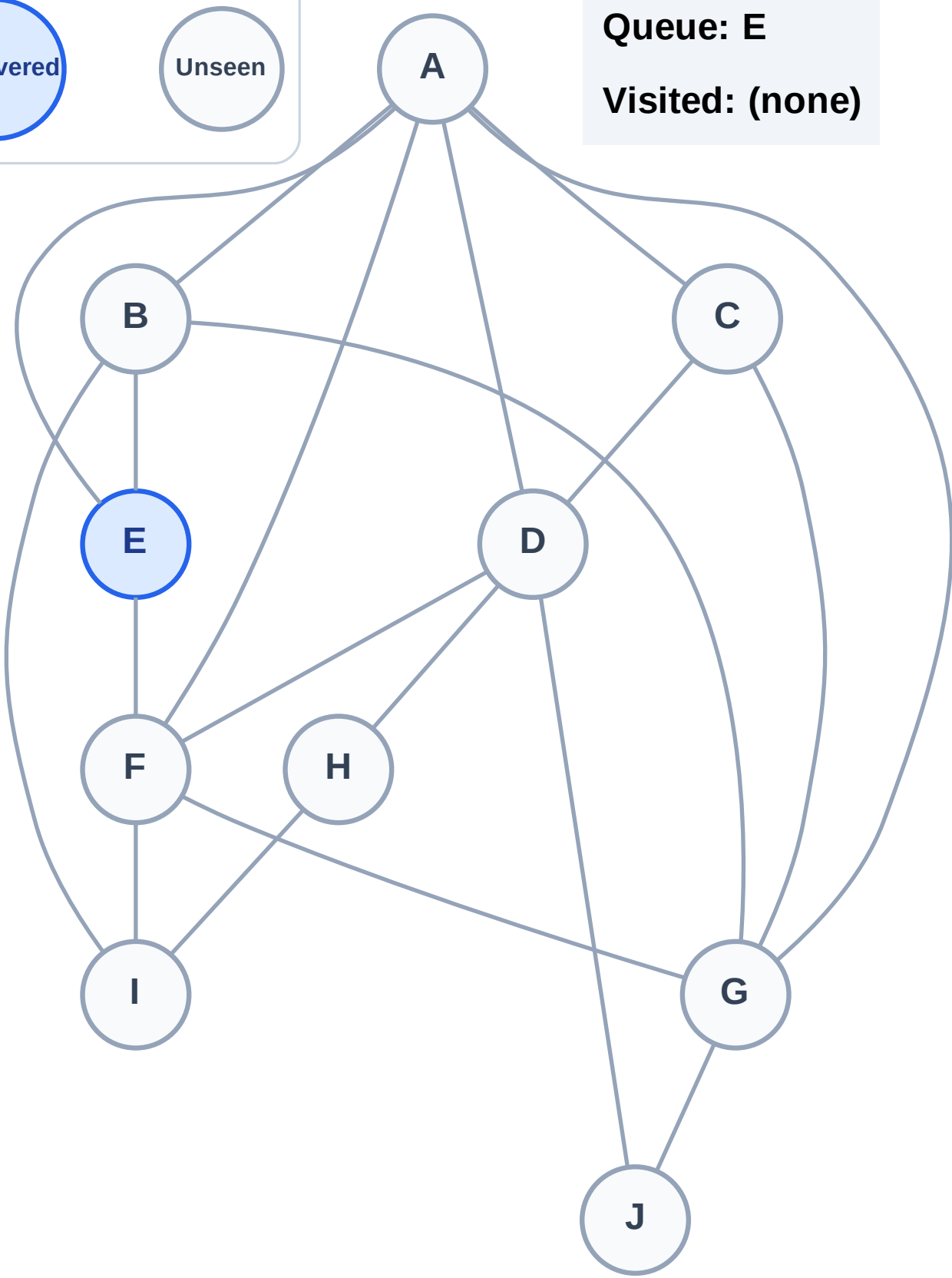
# BFS Traversal

Step 1: Start at E

## Legend



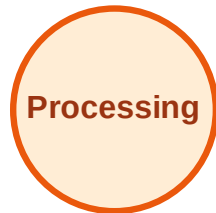
Queue: E  
Visited: (none)



# BFS Traversal

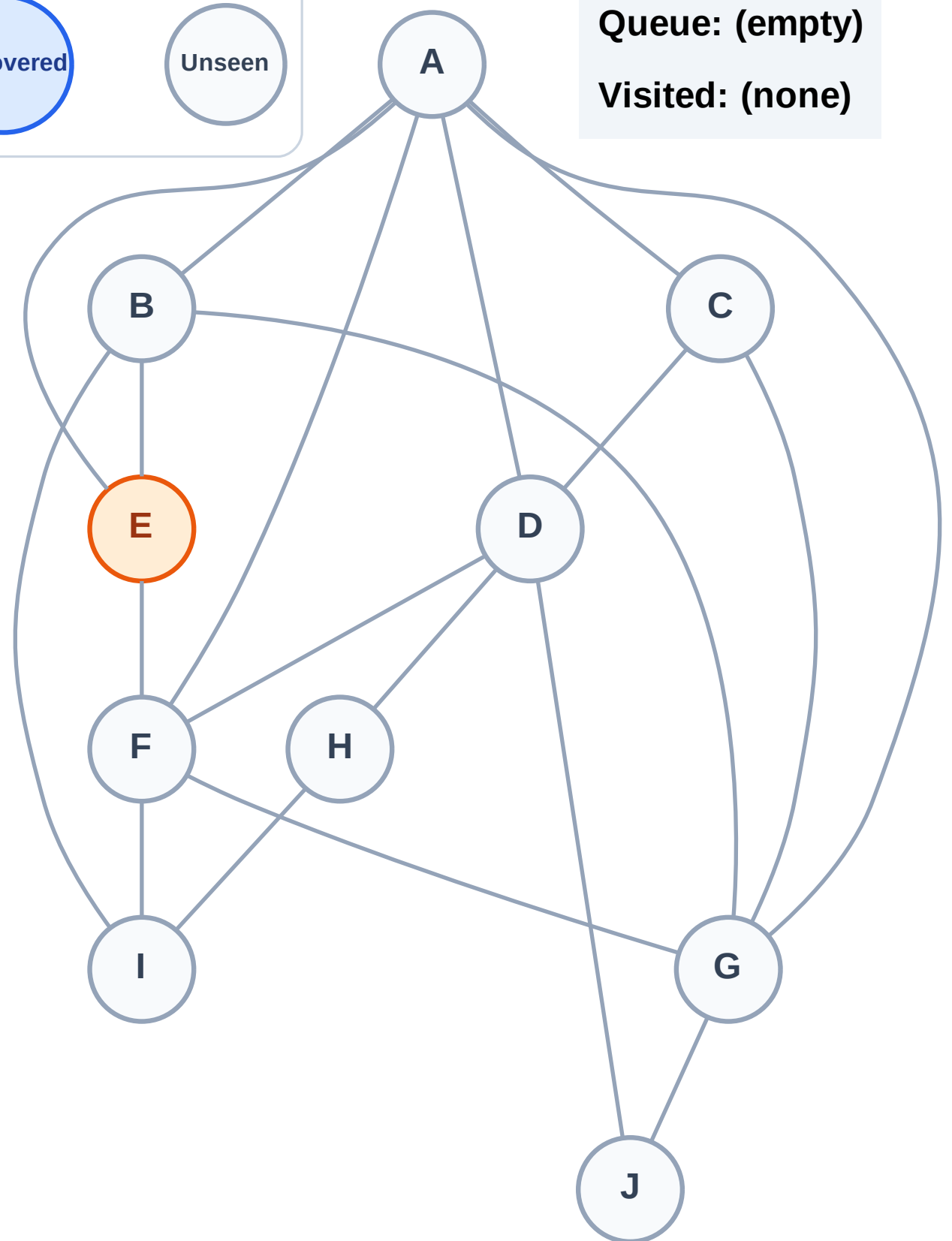
Step 2: Process node E

## Legend



Queue: (empty)

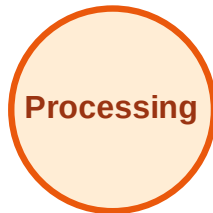
Visited: (none)



# BFS Traversal

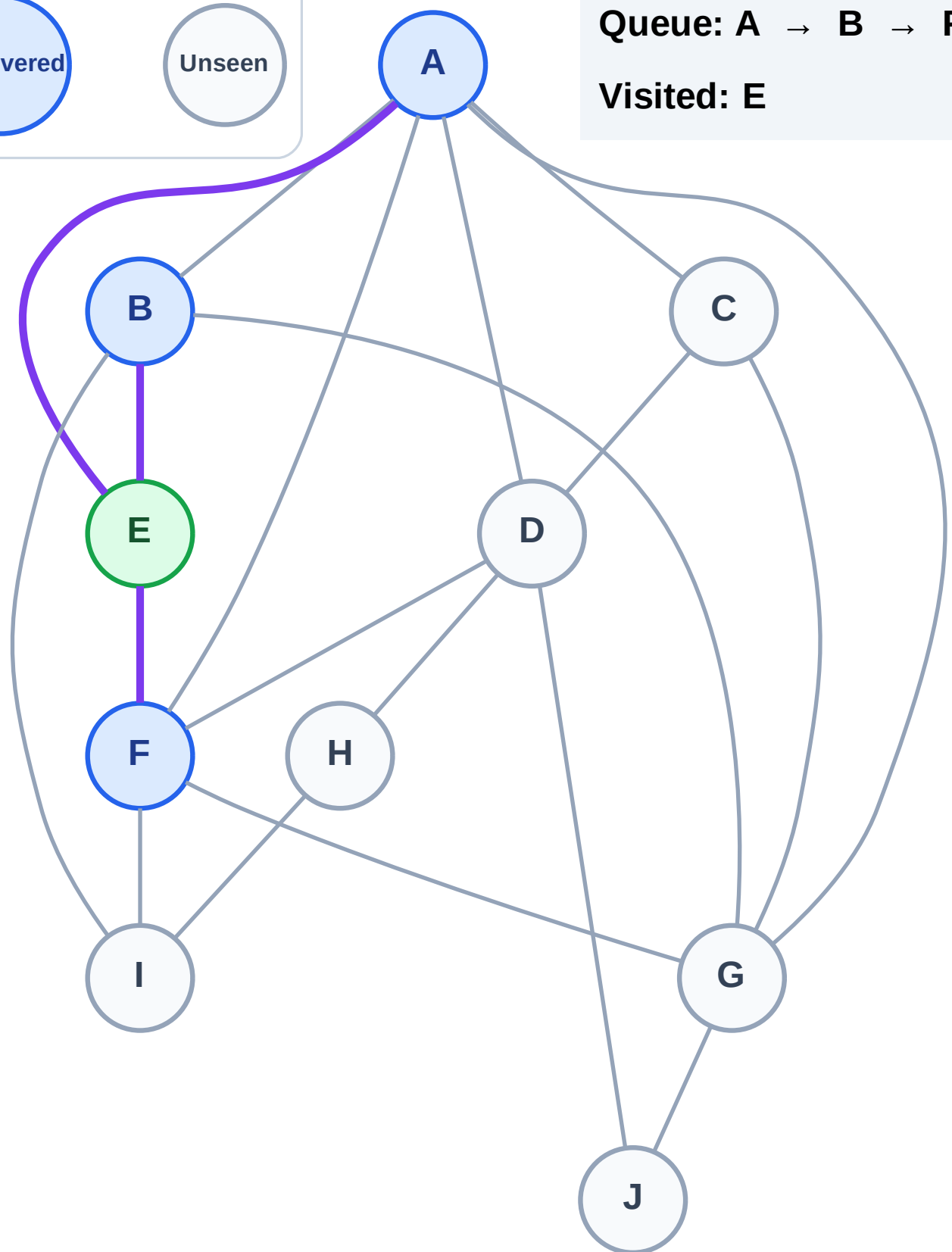
Step 3: Discover A, B, F from E

## Legend



Queue: A → B → F

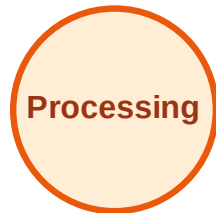
Visited: E



# BFS Traversal

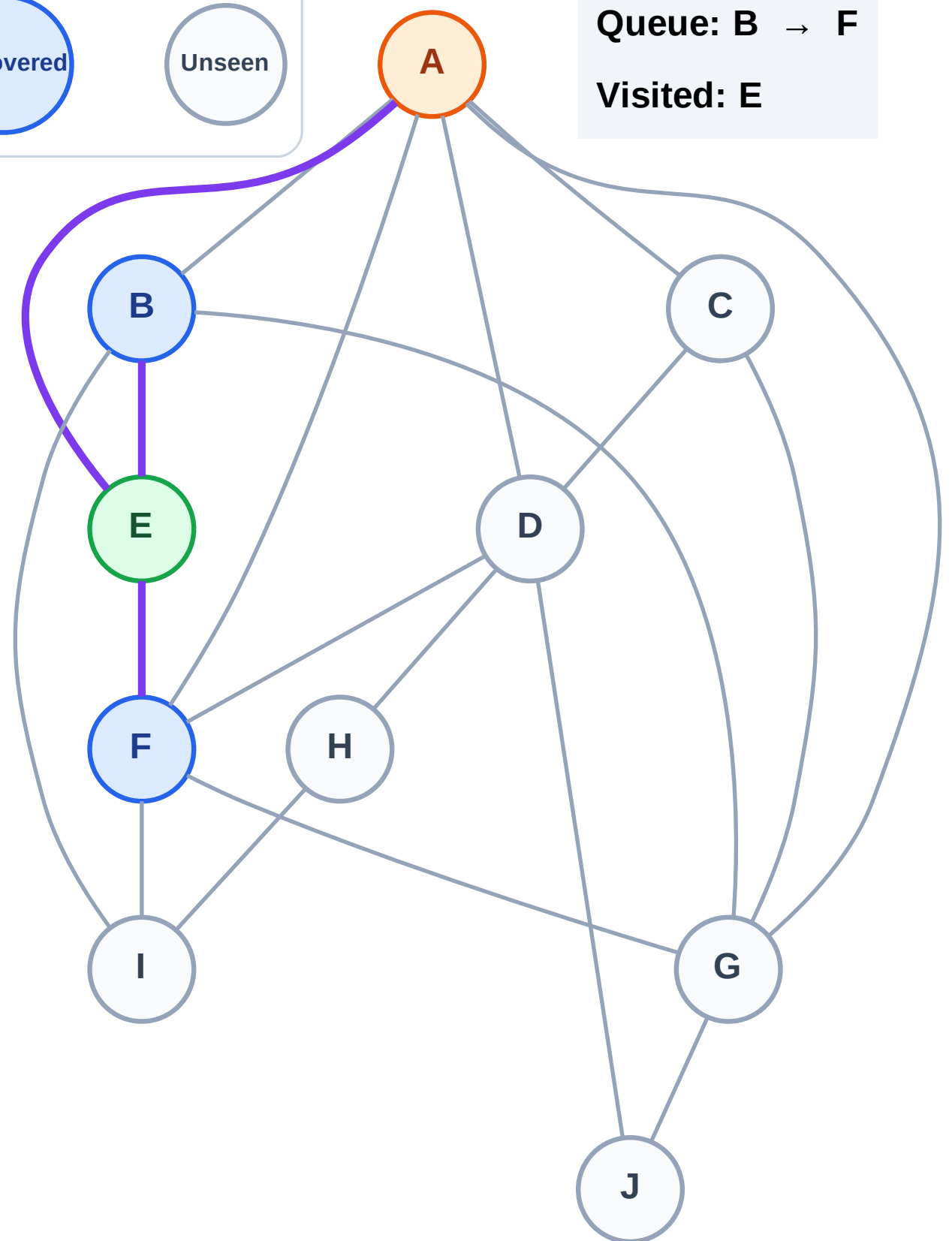
Step 4: Process node A

## Legend



Queue: B → F

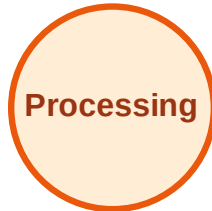
Visited: E



# BFS Traversal

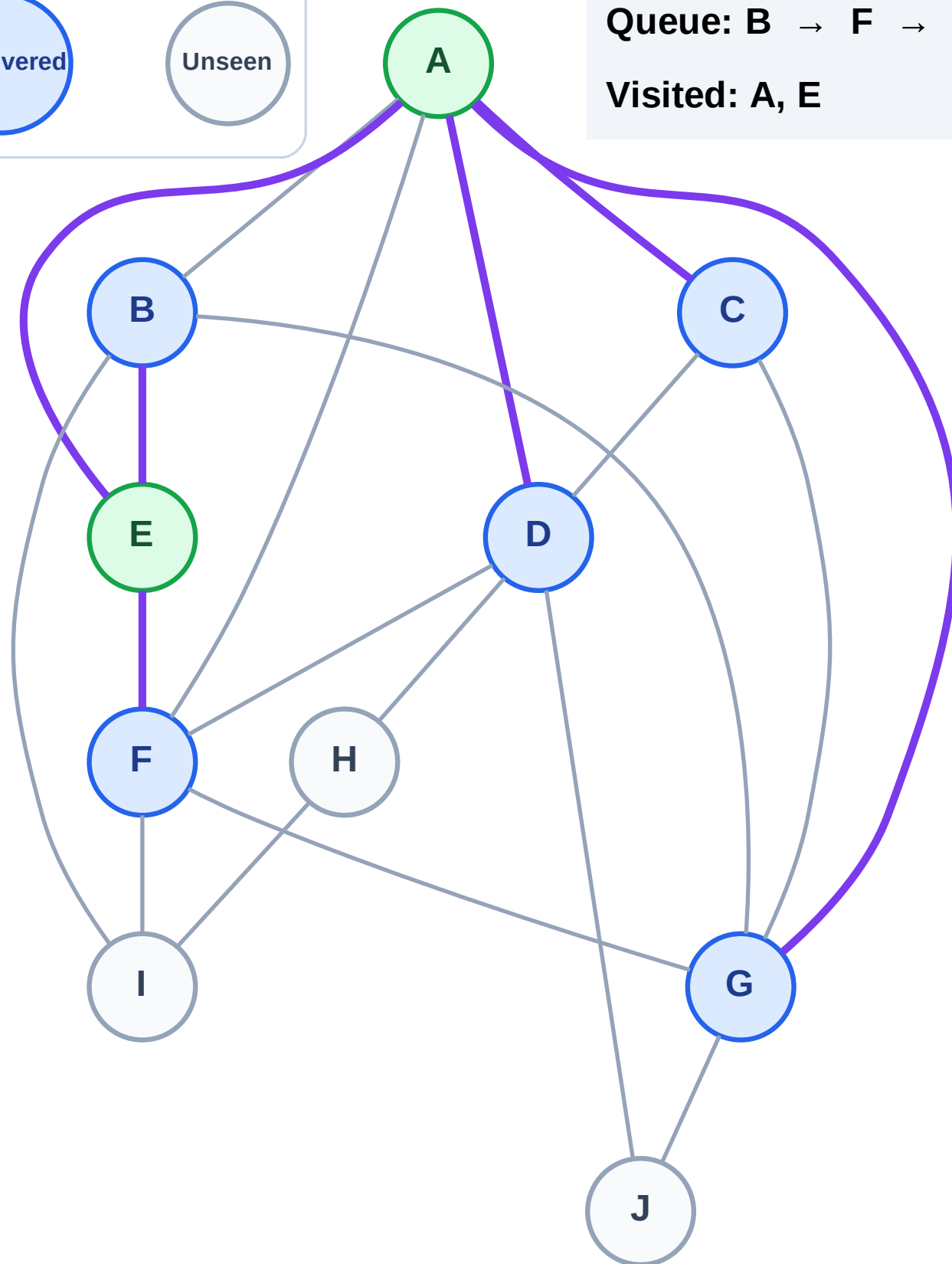
Step 5: Discover C, D, G from A

## Legend



Queue: B → F → C → D → G

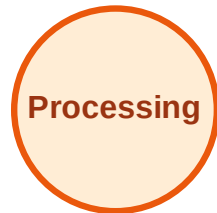
Visited: A, E



# BFS Traversal

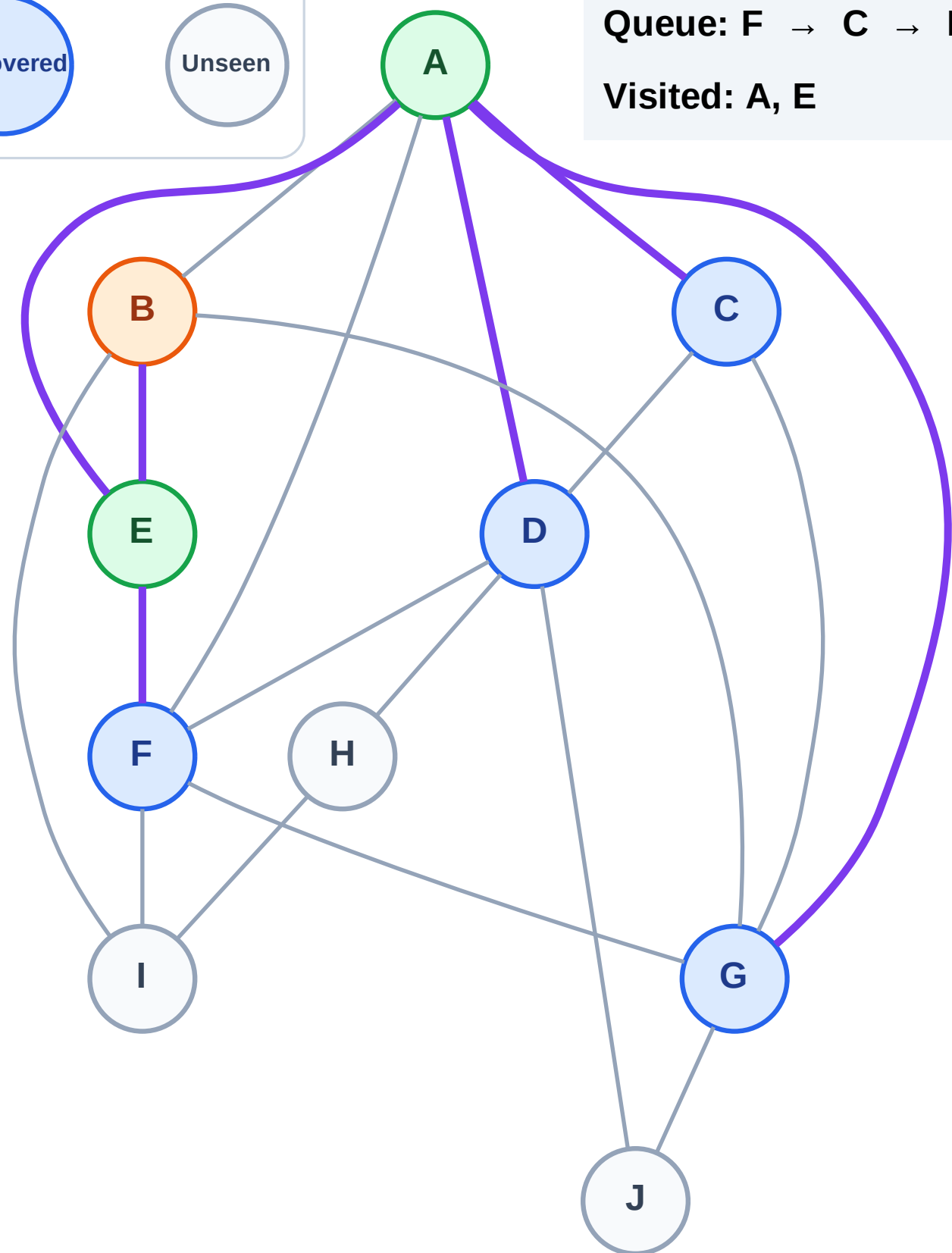
Step 6: Process node B

## Legend



Queue: F → C → D → G

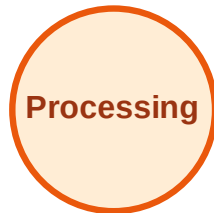
Visited: A, E



# BFS Traversal

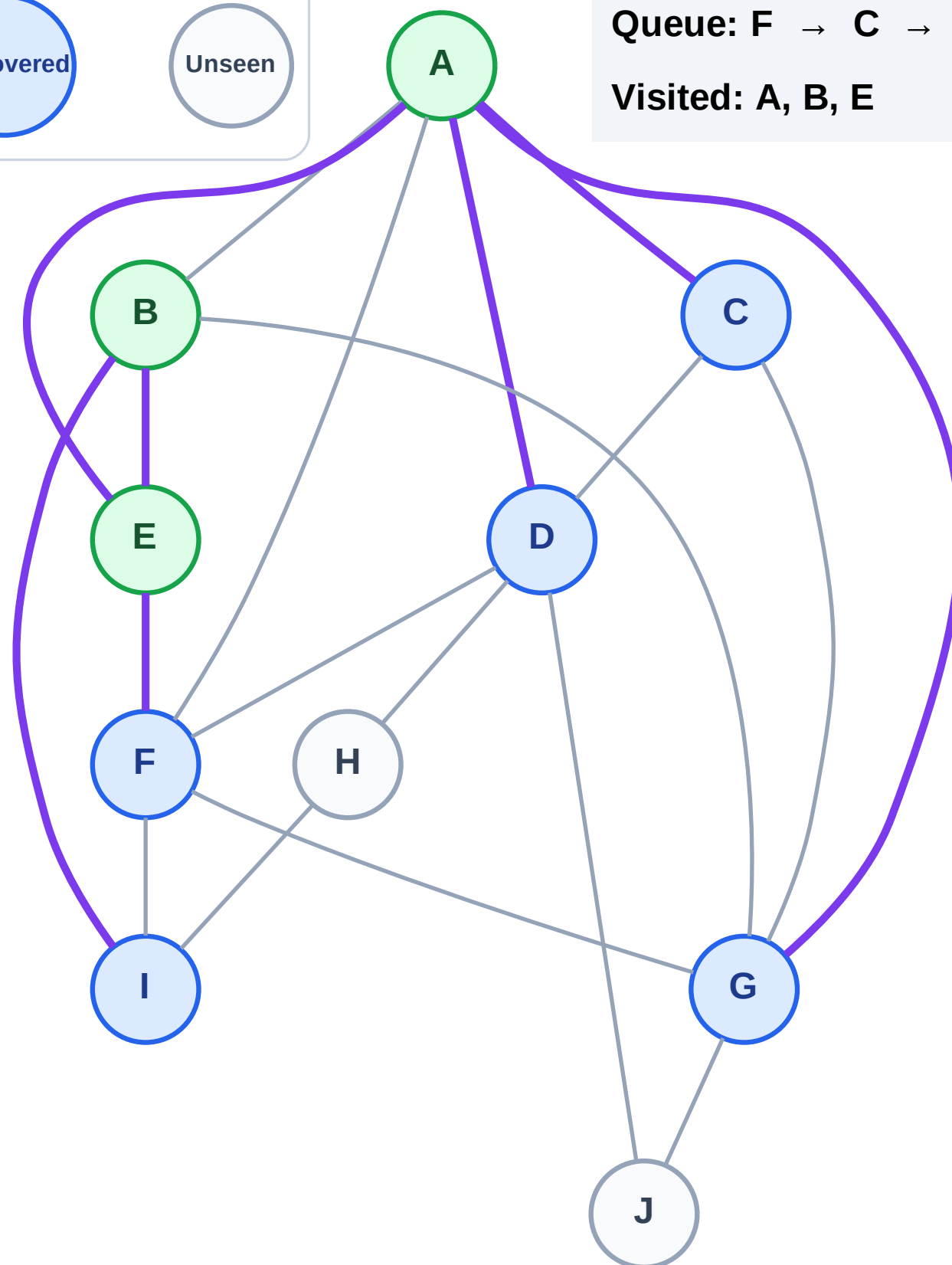
Step 7: Discover I from B

## Legend



Queue: F → C → D → G → I

Visited: A, B, E



# BFS Traversal

Step 8: Process node F

Legend

Complete

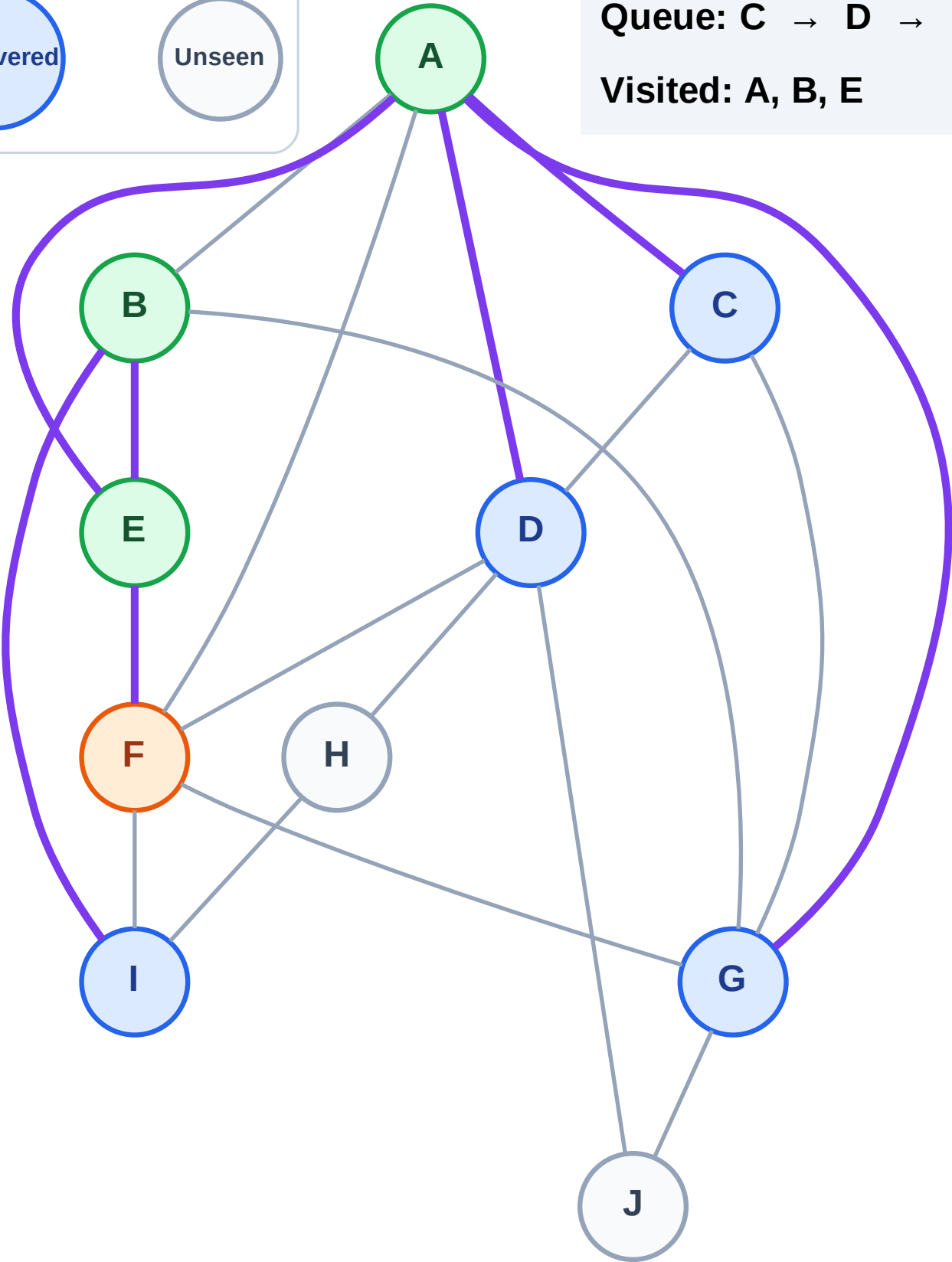
Processing

Discovered

Unseen

Queue: C → D → G → I

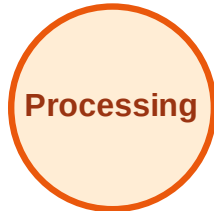
Visited: A, B, E



# BFS Traversal

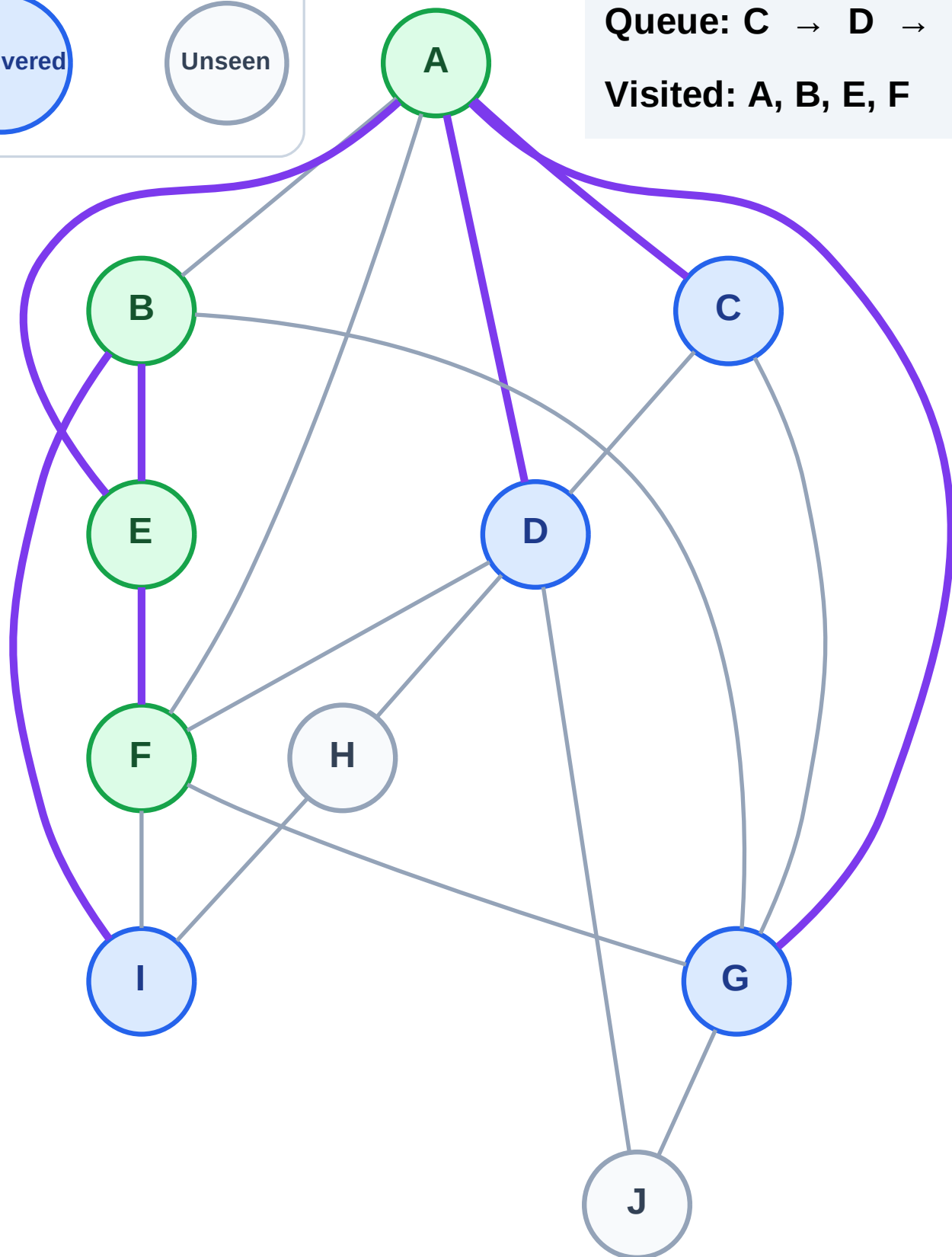
Step 9: No new neighbors from F

## Legend



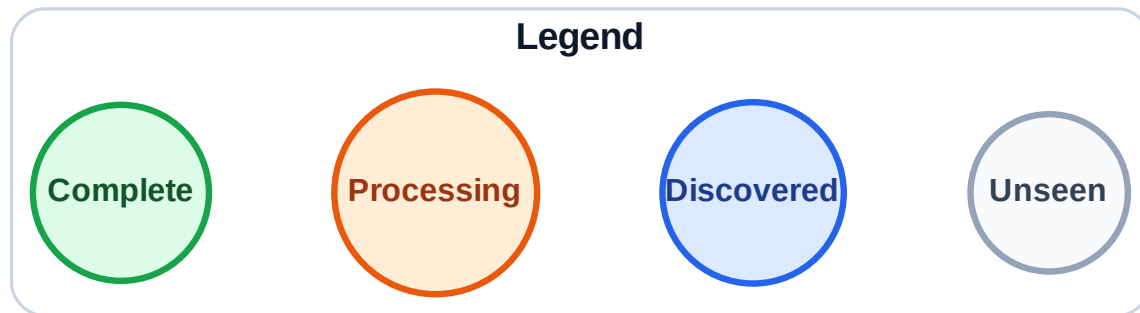
Queue: C → D → G → I

Visited: A, B, E, F

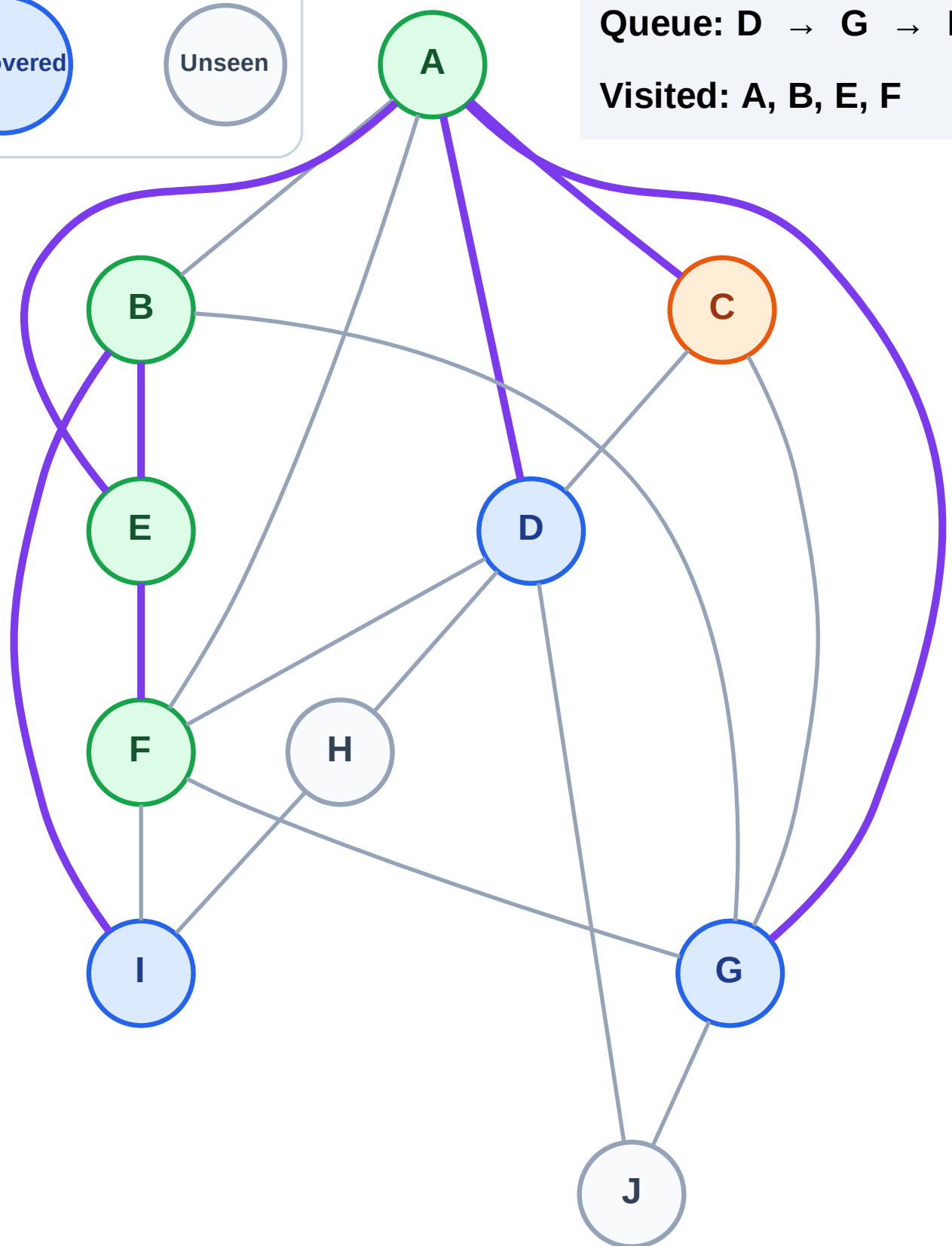


# BFS Traversal

Step 10: Process node C



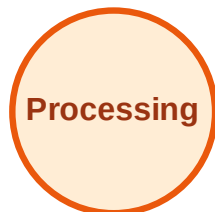
**Queue:** D → G → I  
**Visited:** A, B, E, F



# BFS Traversal

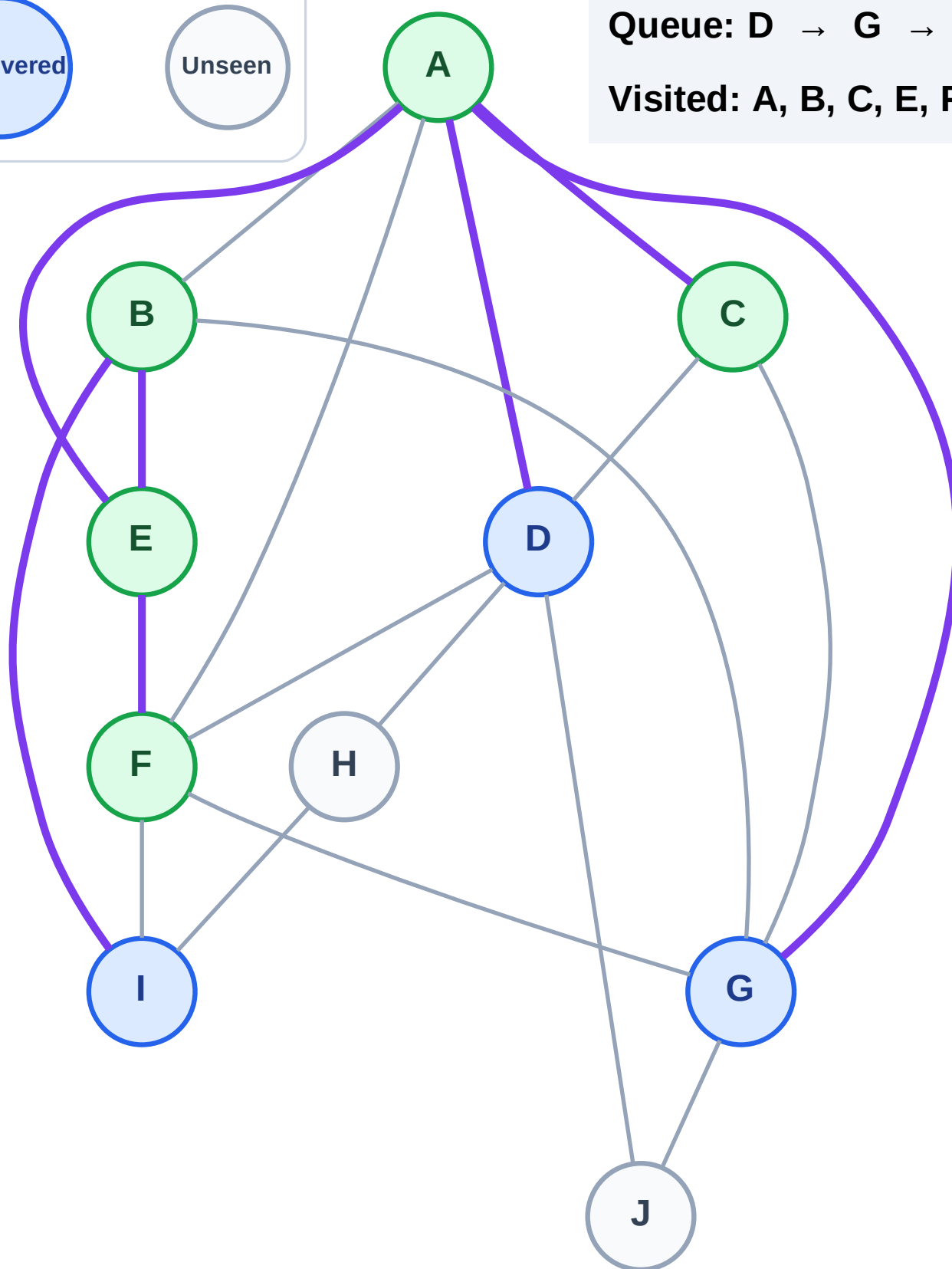
Step 11: No new neighbors from C

## Legend



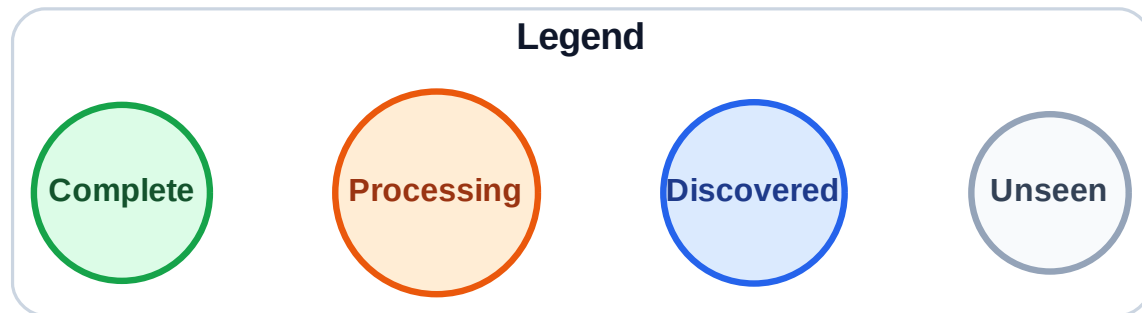
Queue: D → G → I

Visited: A, B, C, E, F

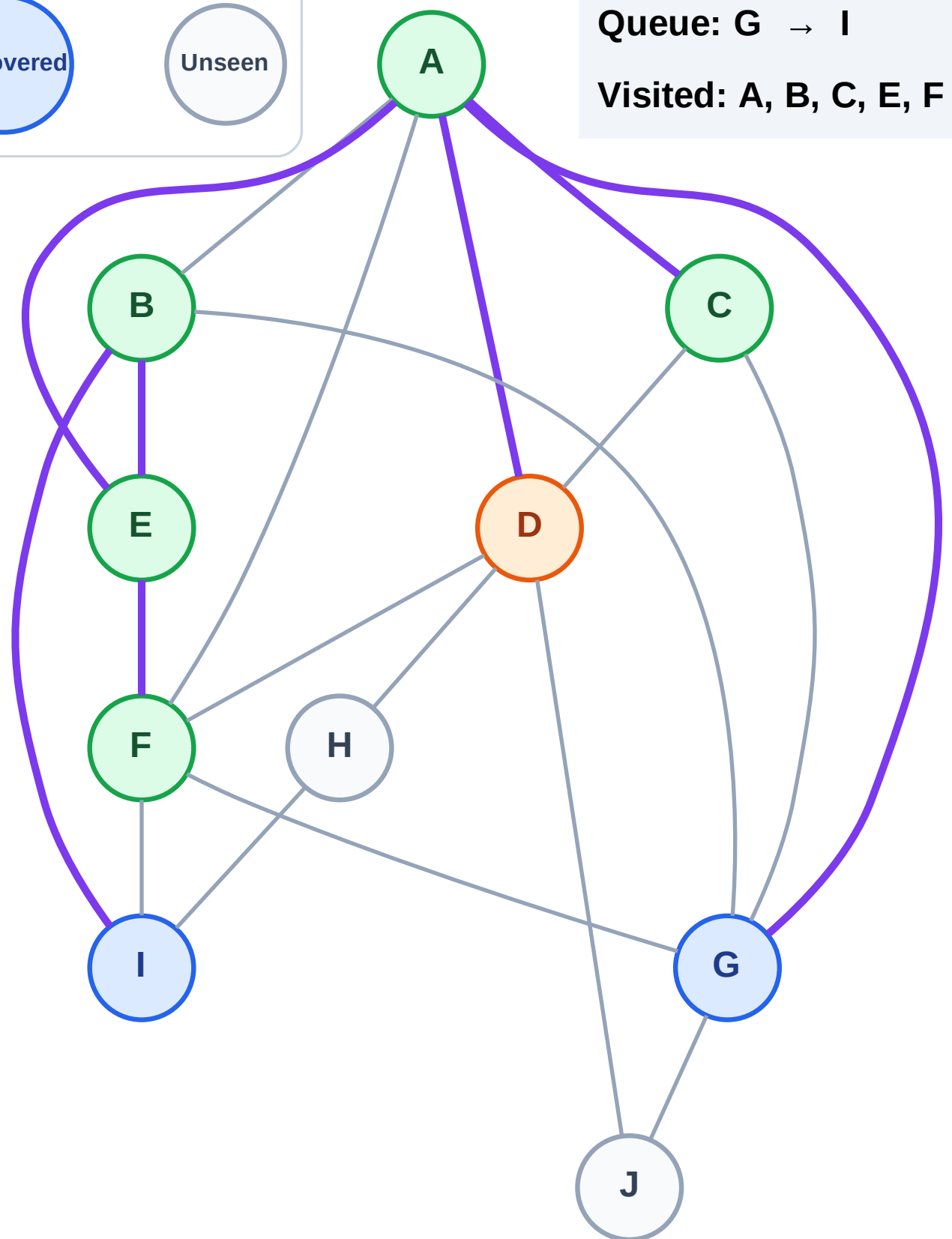


# BFS Traversal

Step 12: Process node D



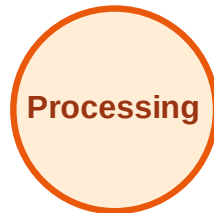
Queue: G → I  
Visited: A, B, C, E, F



# BFS Traversal

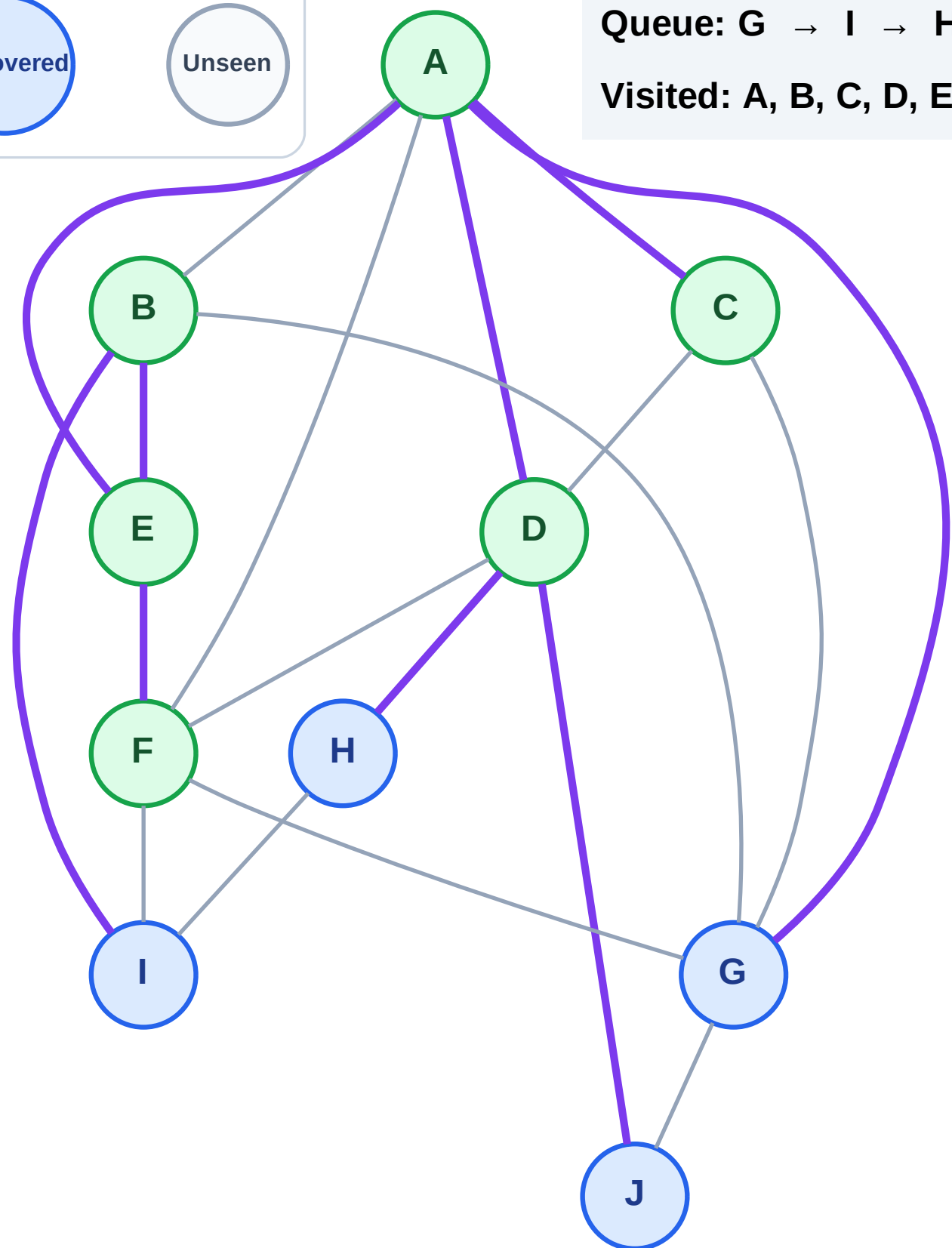
Step 13: Discover H, J from D

## Legend



Queue: G → I → H → J

Visited: A, B, C, D, E, F



# BFS Traversal

Step 14: Process node G

Legend

Complete

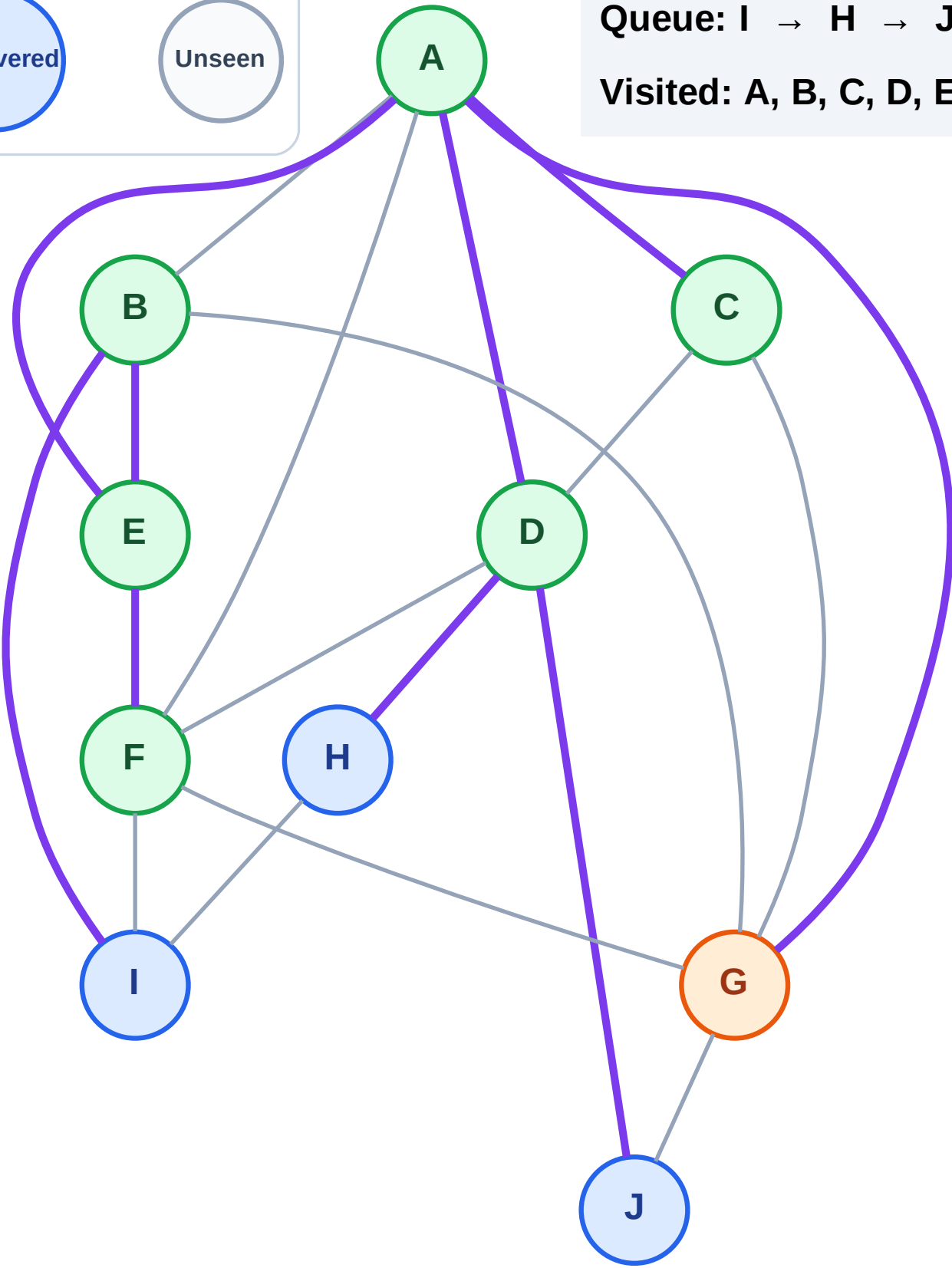
Processing

Discovered

Unseen

Queue: I → H → J

Visited: A, B, C, D, E, F



# BFS Traversal

Step 15: No new neighbors from G

Legend

Complete

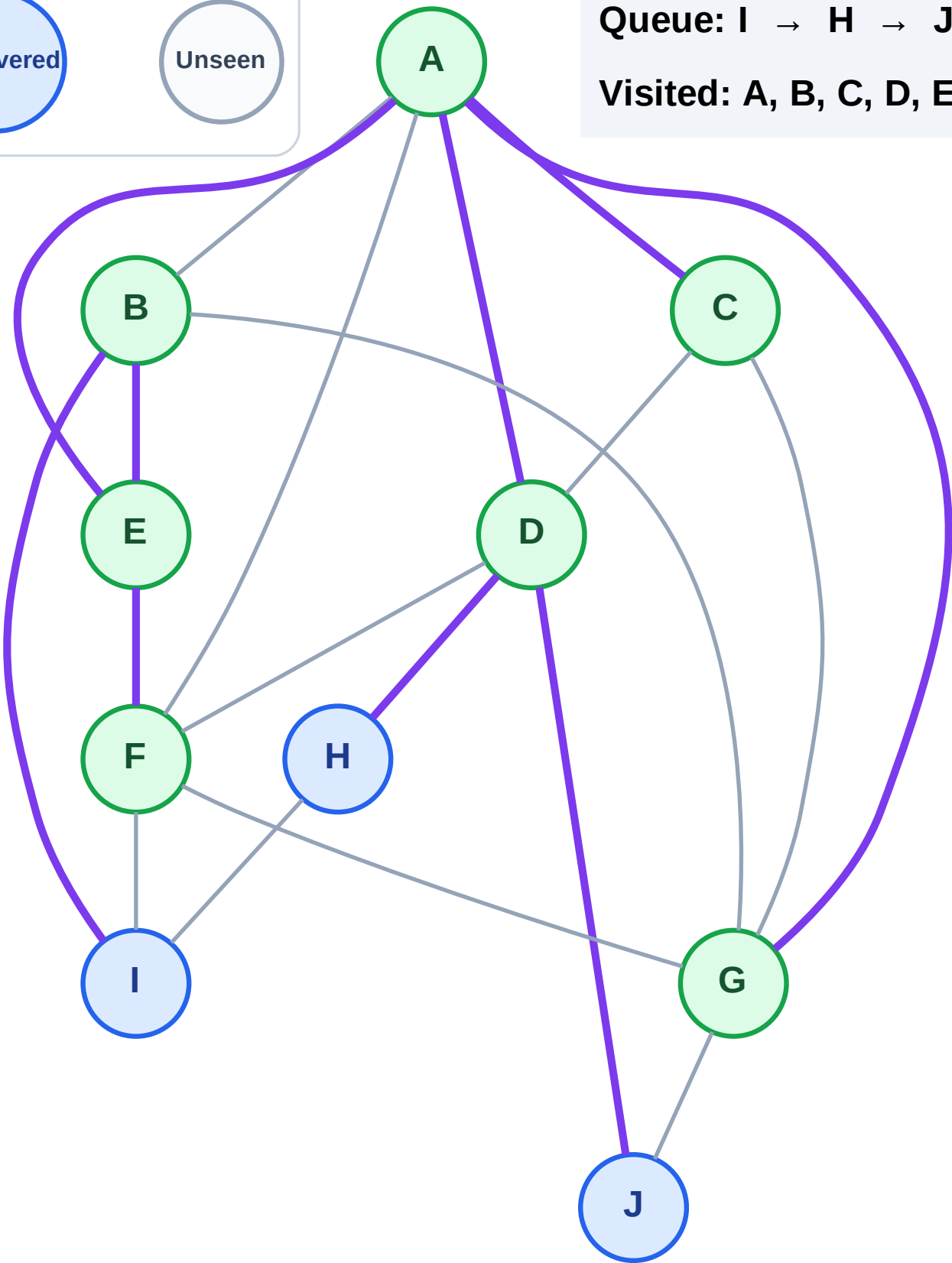
Processing

Discovered

Unseen

Queue: I → H → J

Visited: A, B, C, D, E, F, G



# BFS Traversal

Step 16: Process node I

Legend

Complete

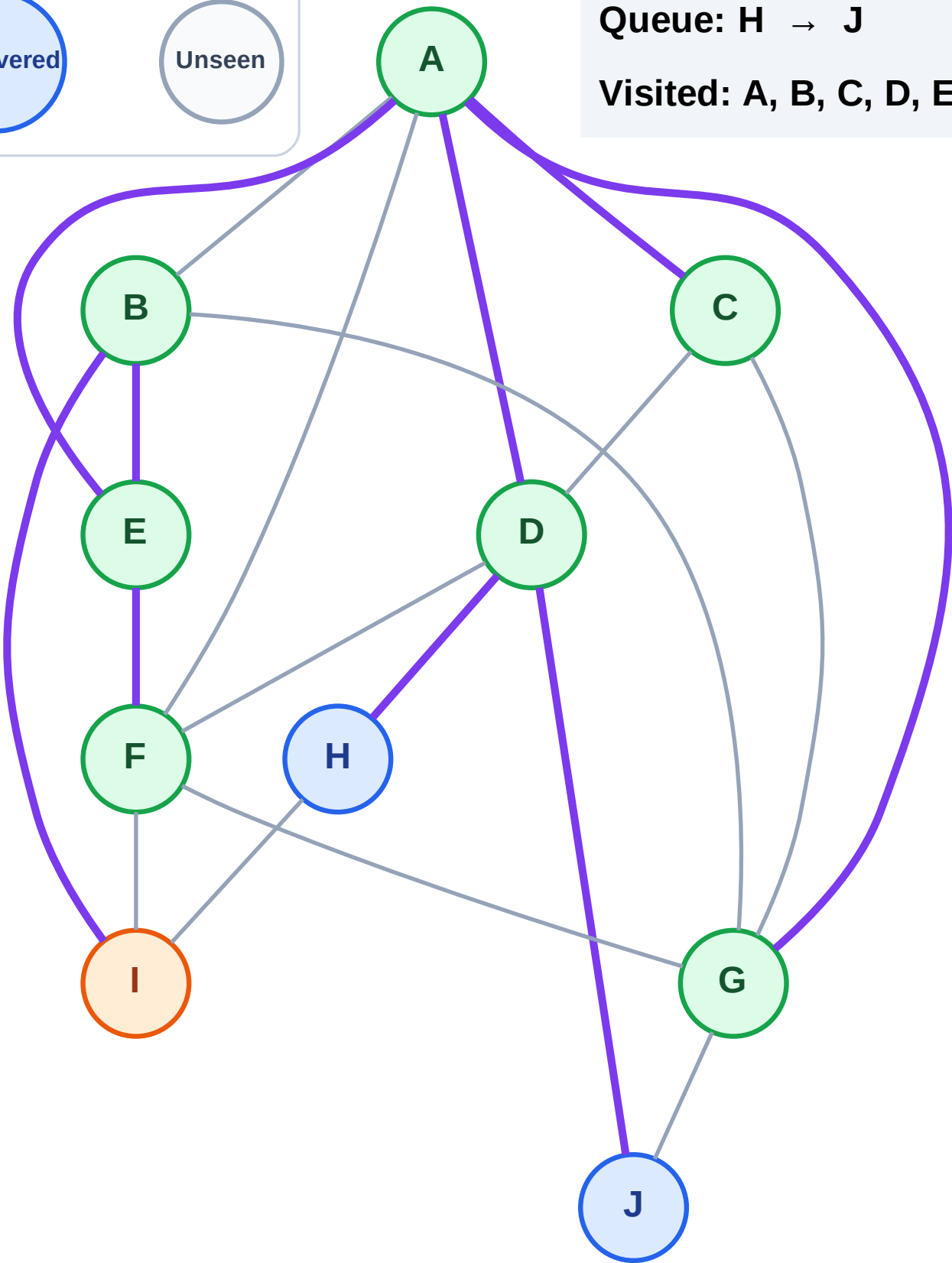
Processing

Discovered

Unseen

Queue: H → J

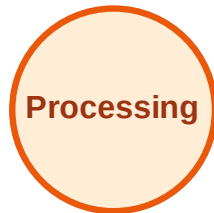
Visited: A, B, C, D, E, F, G



# BFS Traversal

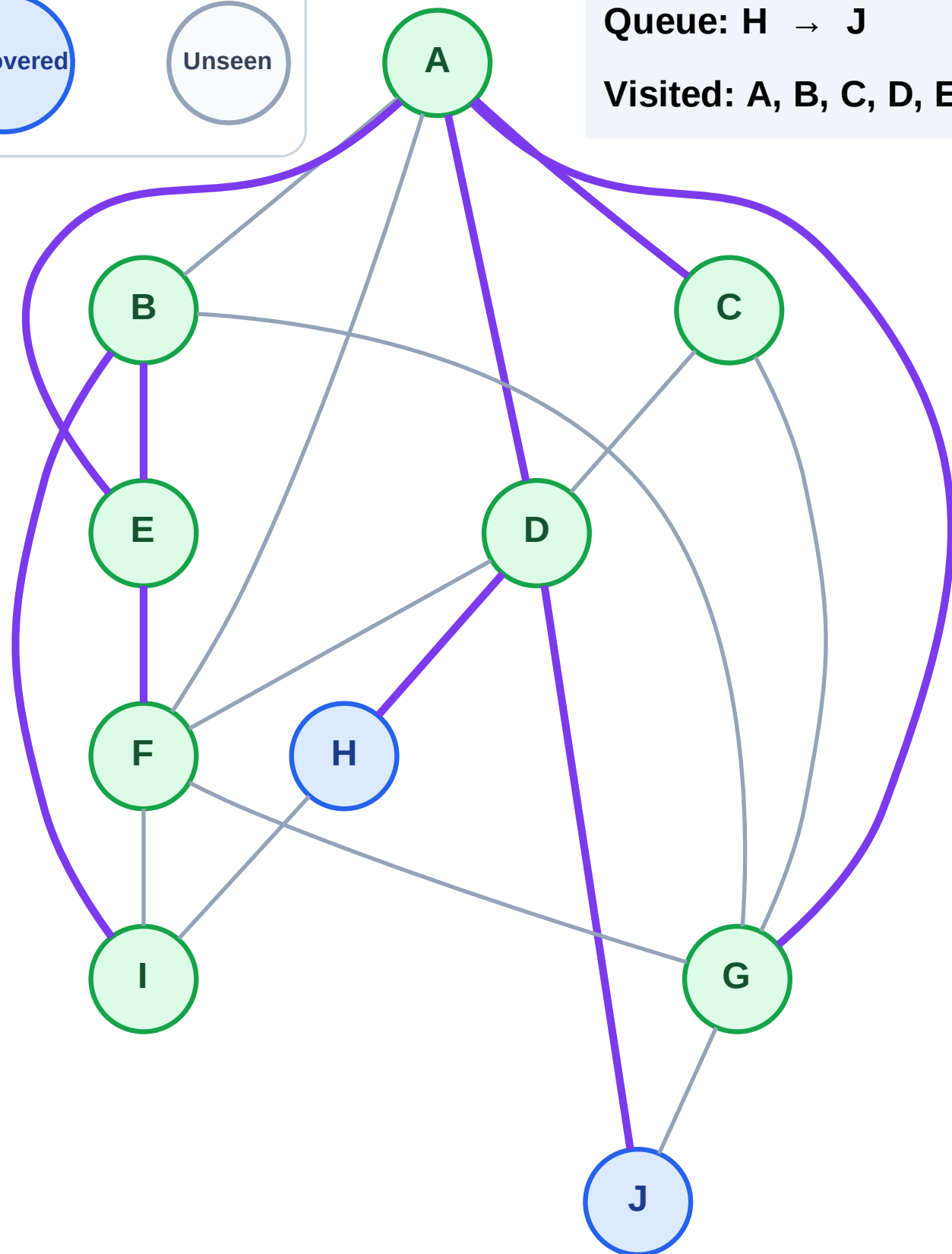
Step 17: No new neighbors from I

## Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



# BFS Traversal

Step 18: Process node H

Legend

Complete

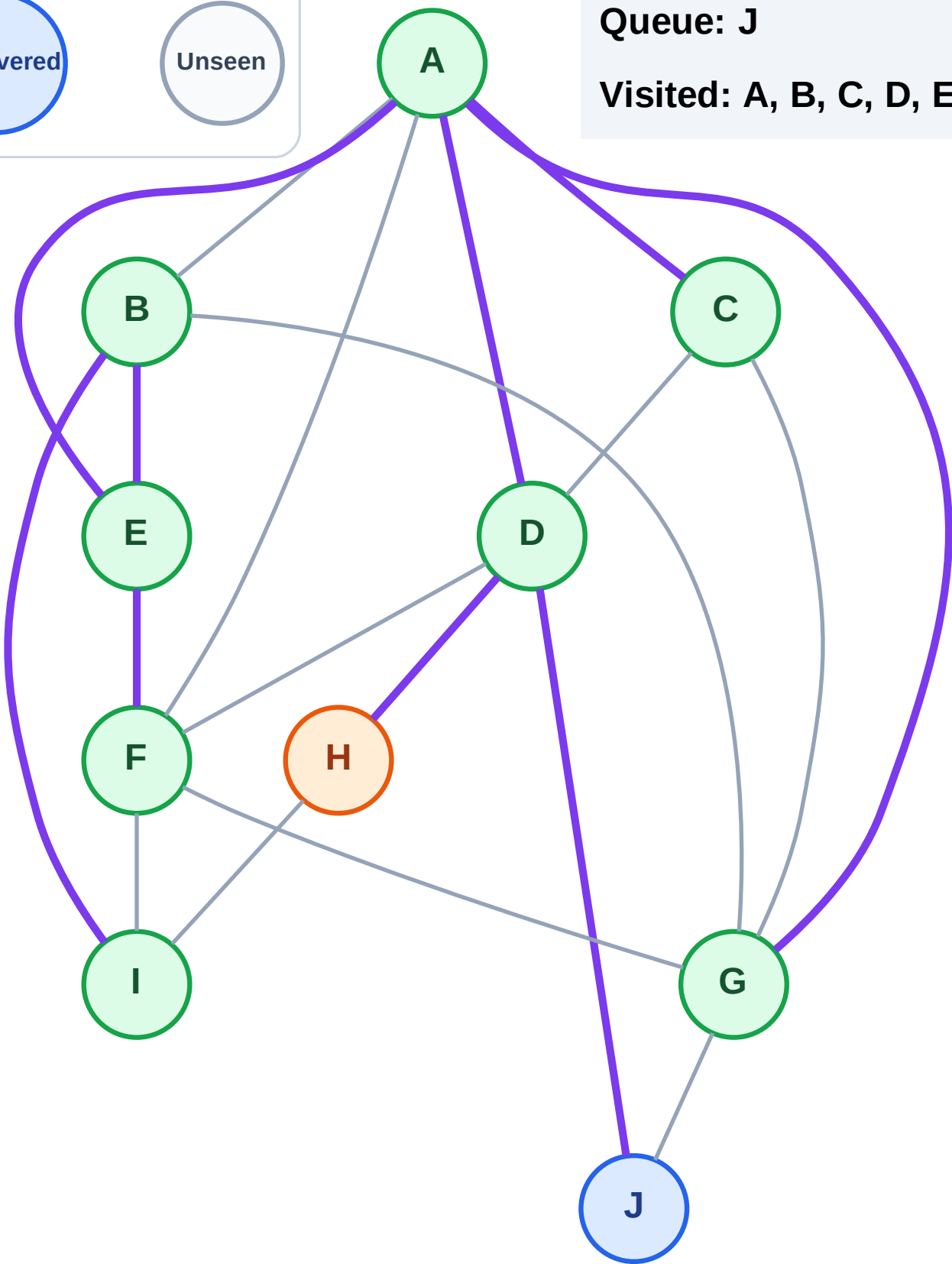
Processing

Discovered

Unseen

Queue: J

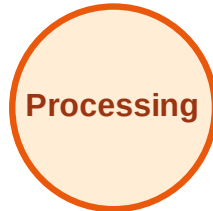
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

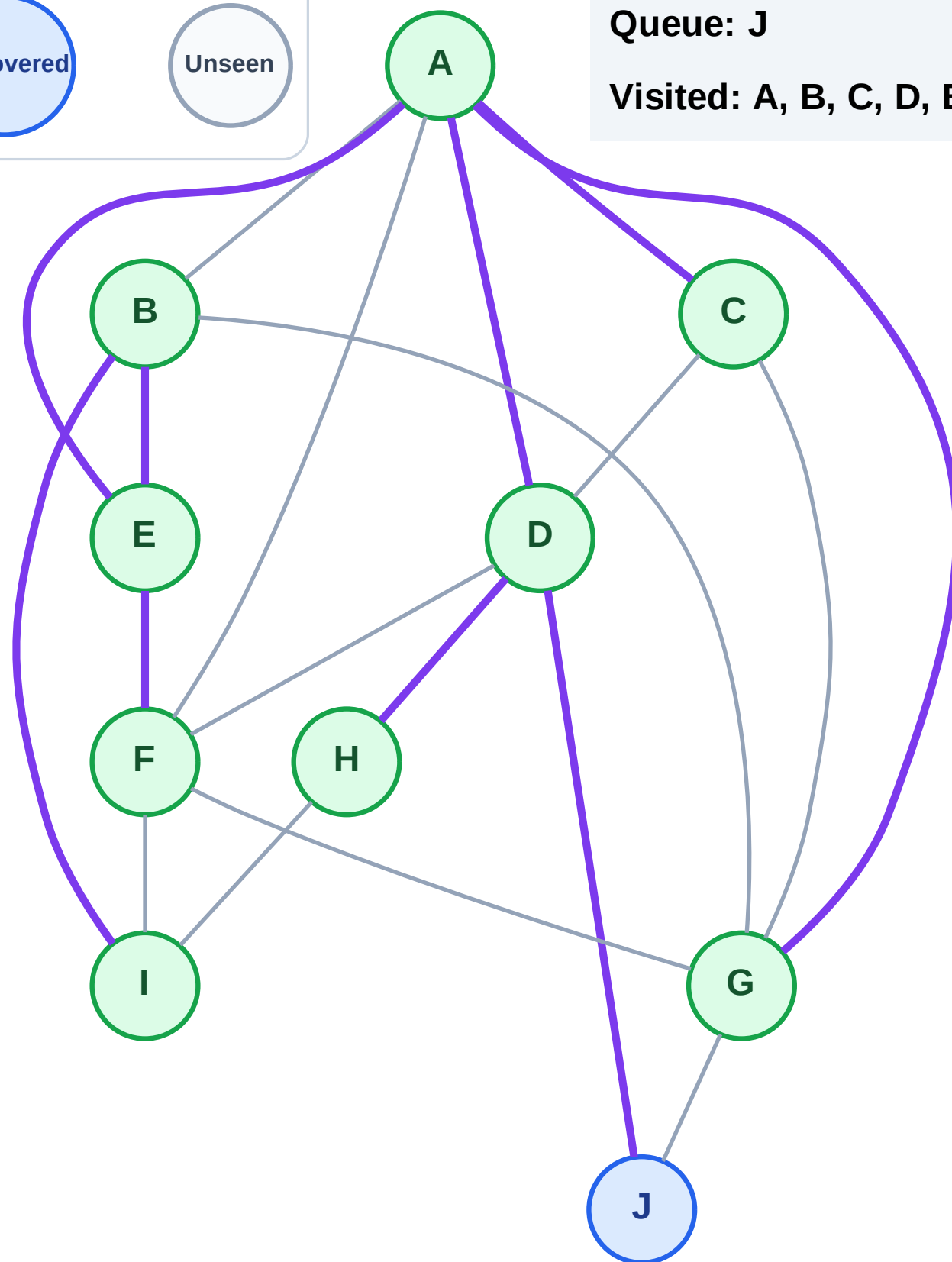
Step 19: No new neighbors from H

## Legend



Queue: J

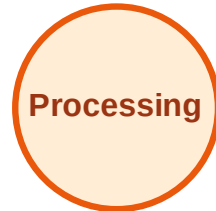
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

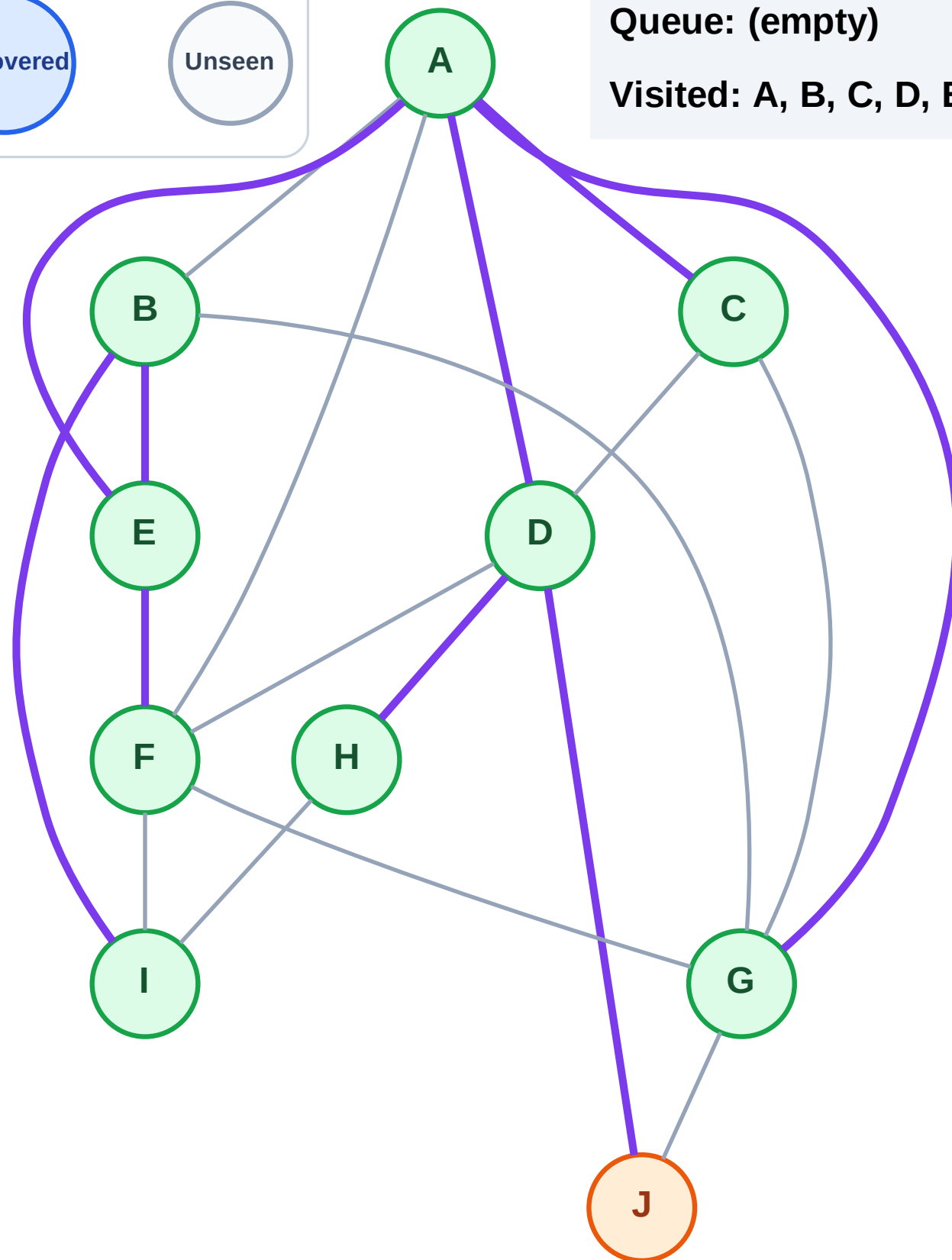
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

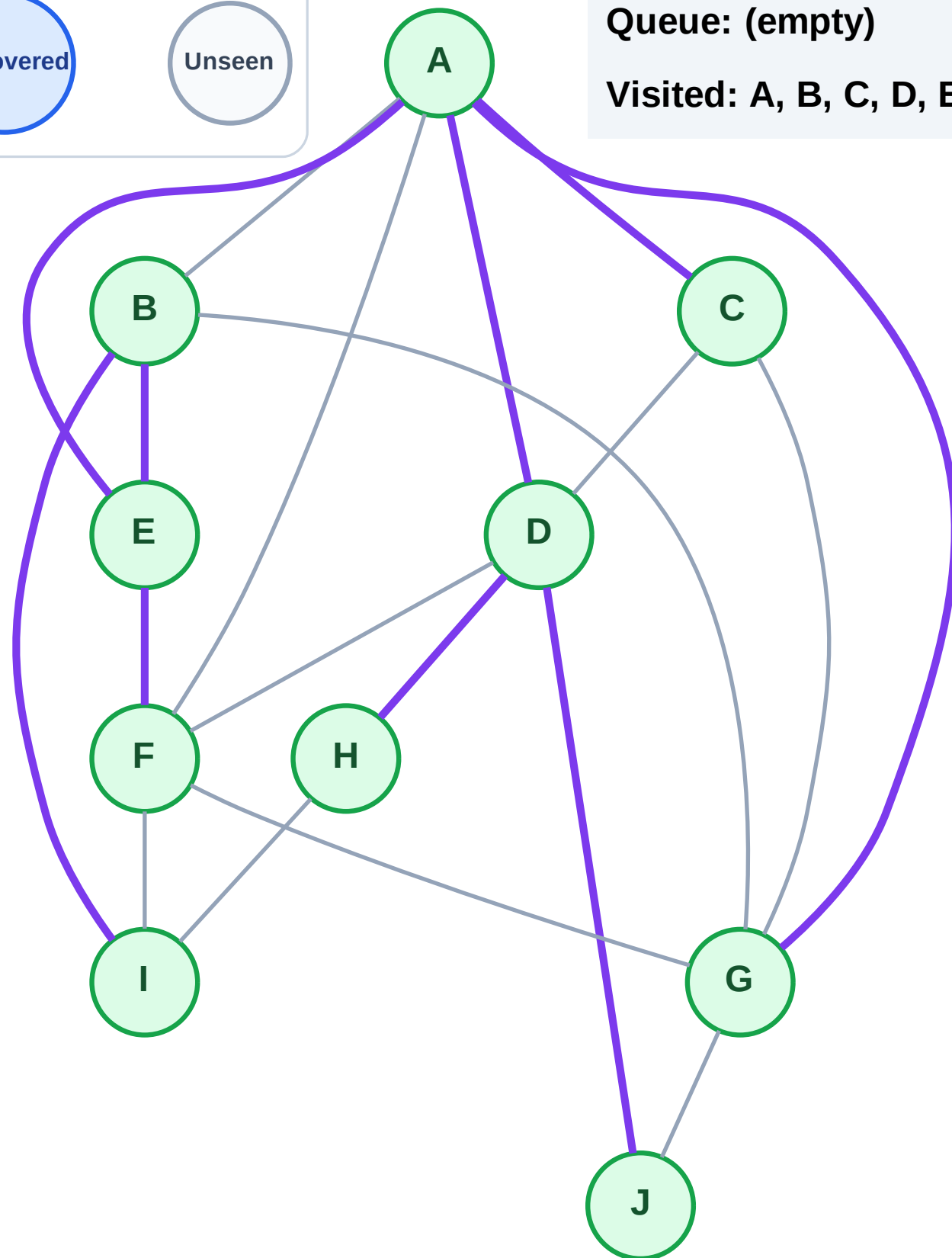
Step 21: No new neighbors from J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

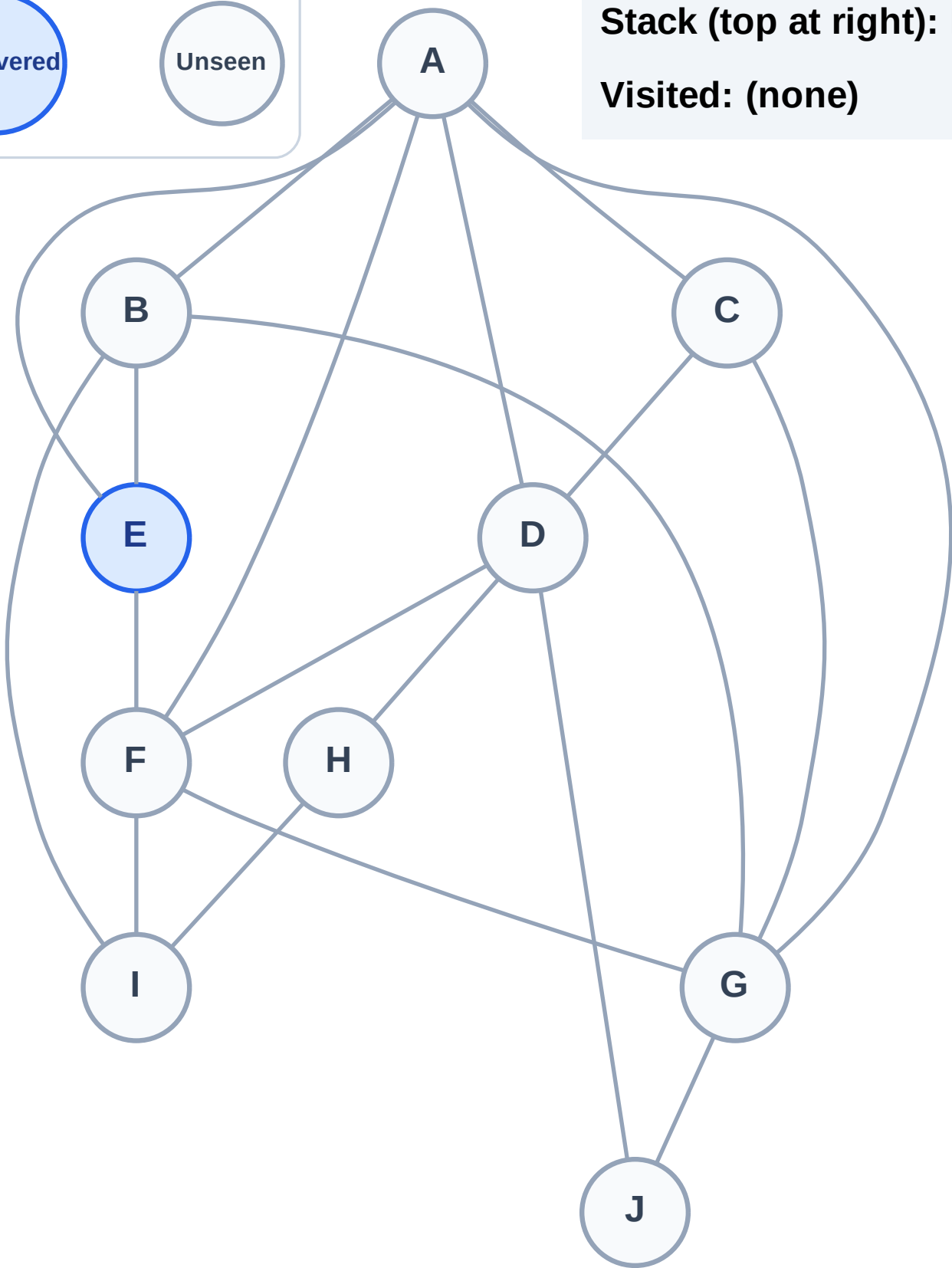
Step 1: Start at E

## Legend



Stack (top at right): E

Visited: (none)



# DFS Traversal

Step 2: Process node E

Legend

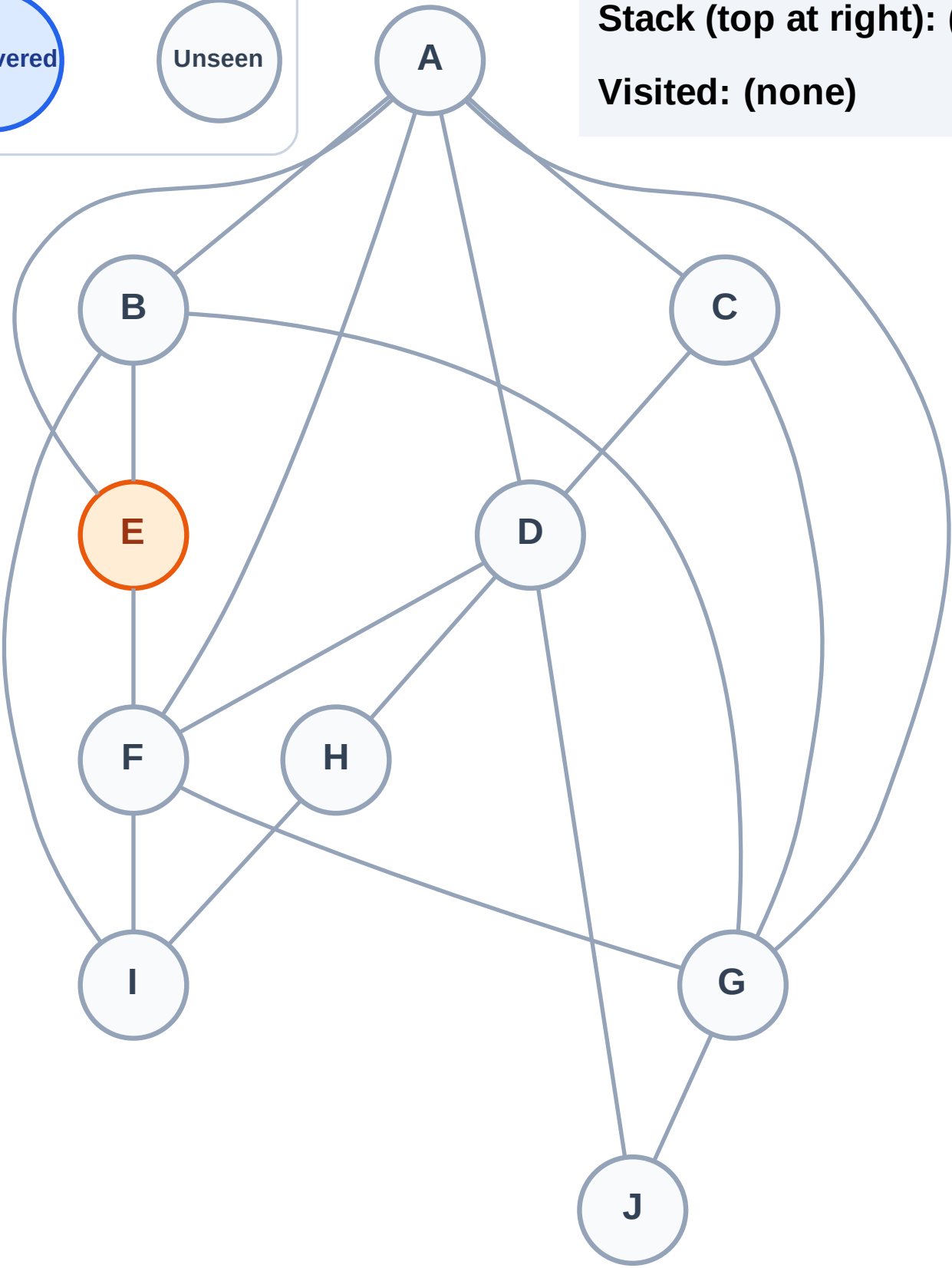
Complete

Processing

Discovered

Unseen

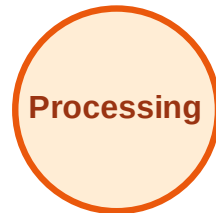
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

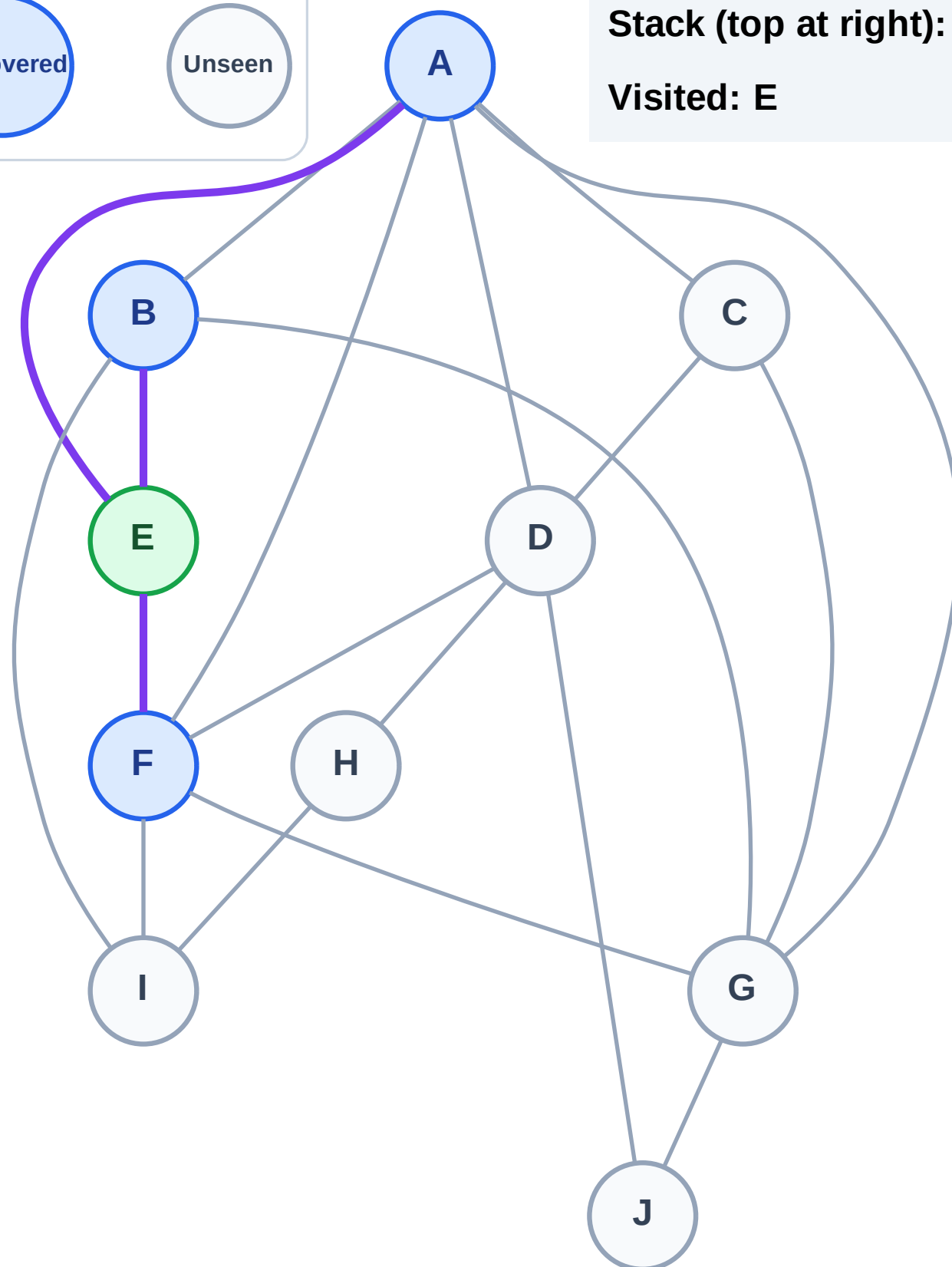
Step 3: Discover F, B, A from E

## Legend



Stack (top at right): F | B | A

Visited: E



# DFS Traversal

Step 4: Process node A

Legend

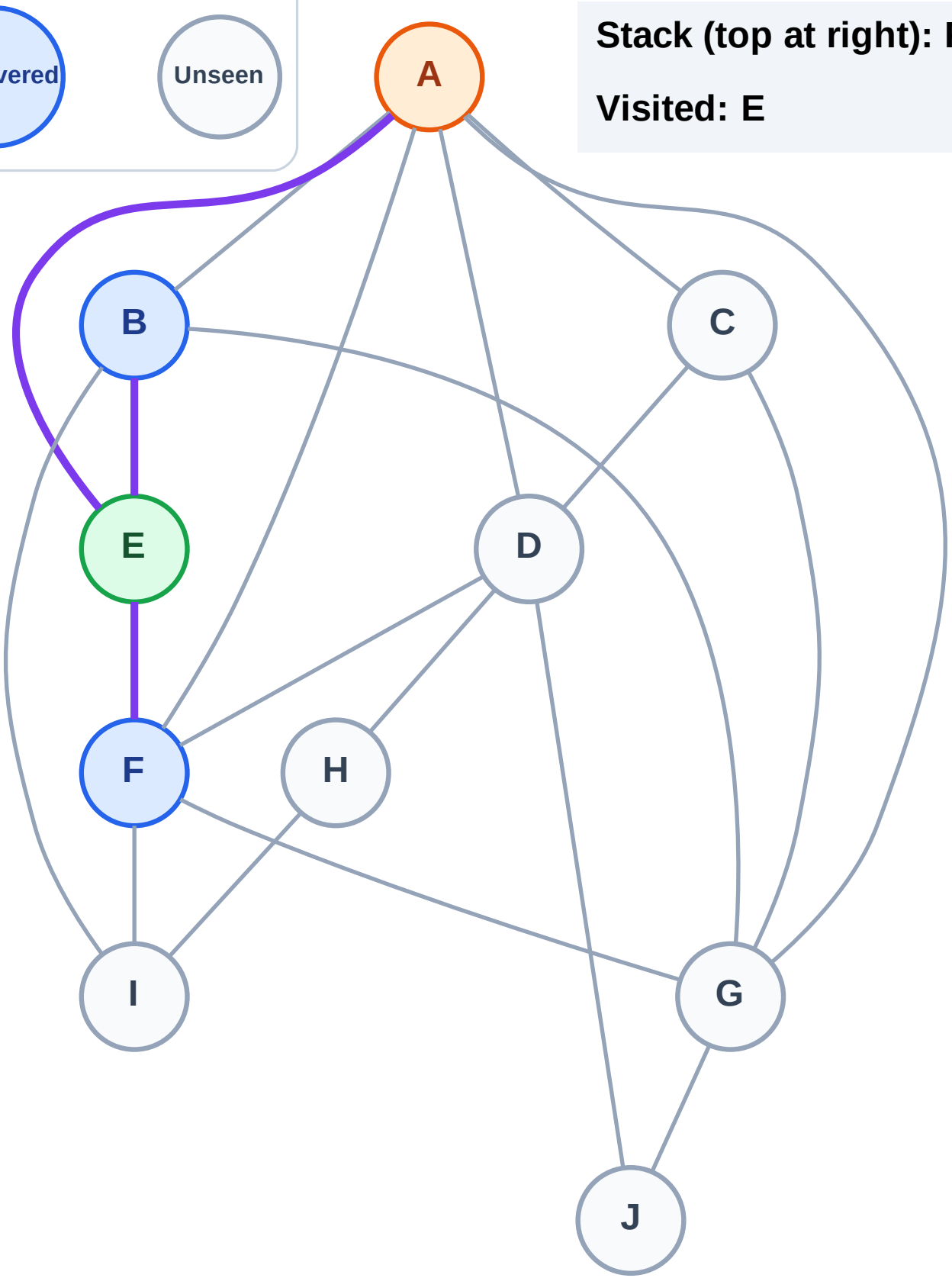
Complete

Processing

Discovered

Unseen

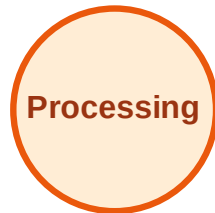
Stack (top at right): F | B  
Visited: E



# DFS Traversal

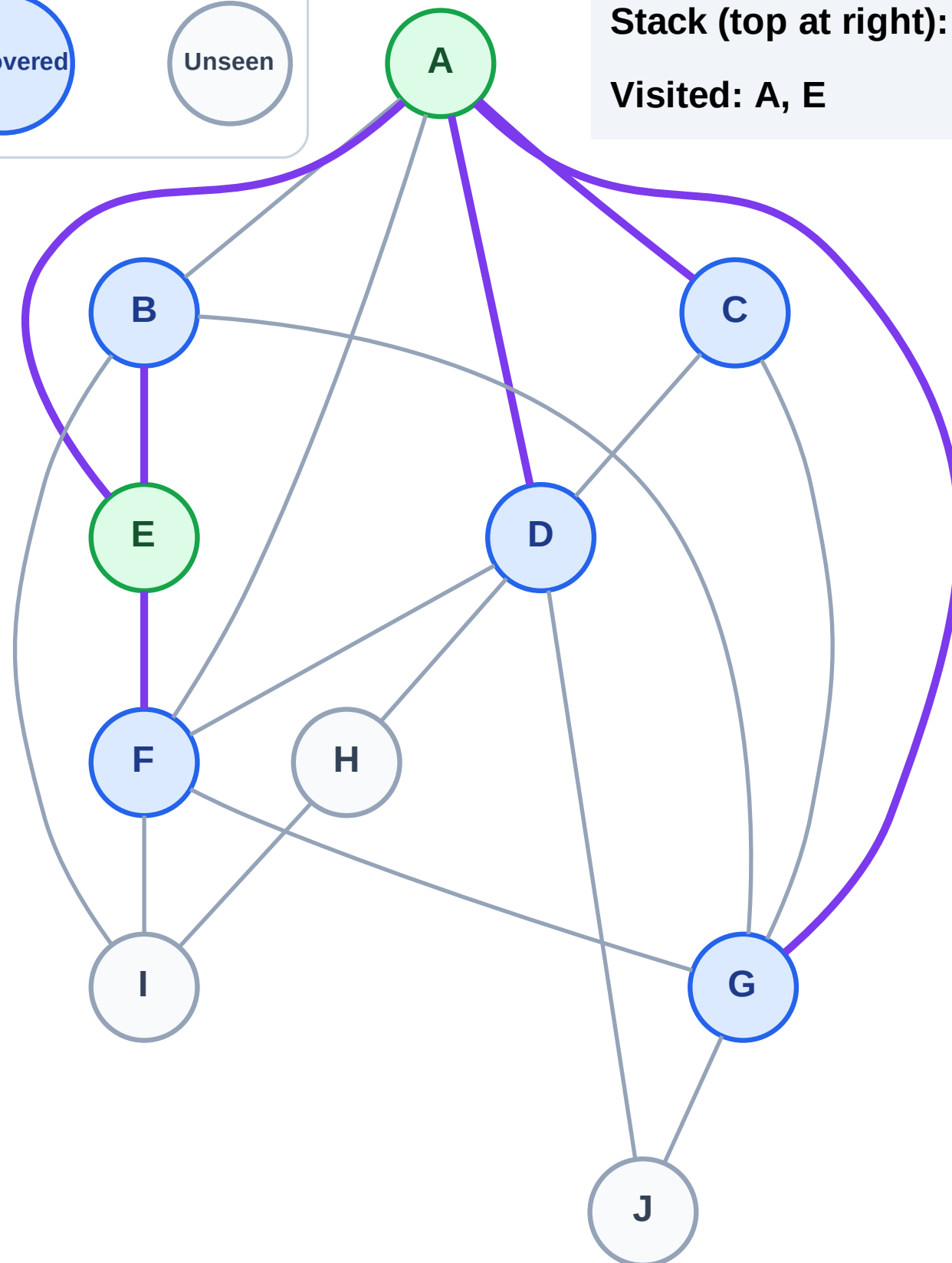
Step 5: Discover G, D, C from A

## Legend



Stack (top at right): F | B | G | D | C

Visited: A, E



# DFS Traversal

Step 6: Process node C

Legend

Complete

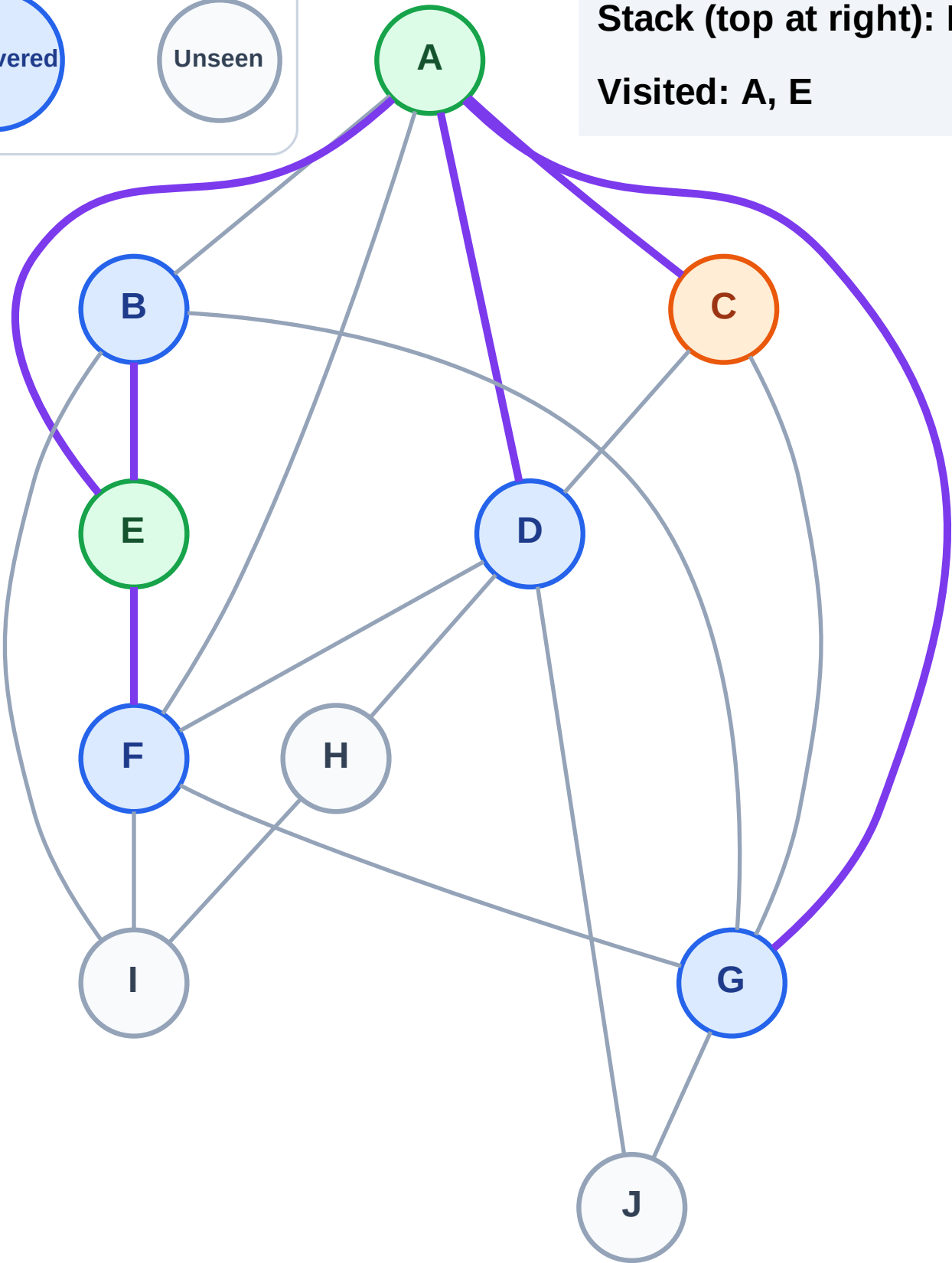
Processing

Discovered

Unseen

Stack (top at right): F | B | G | D

Visited: A, E



# DFS Traversal

Step 7: No new neighbors from C

Legend

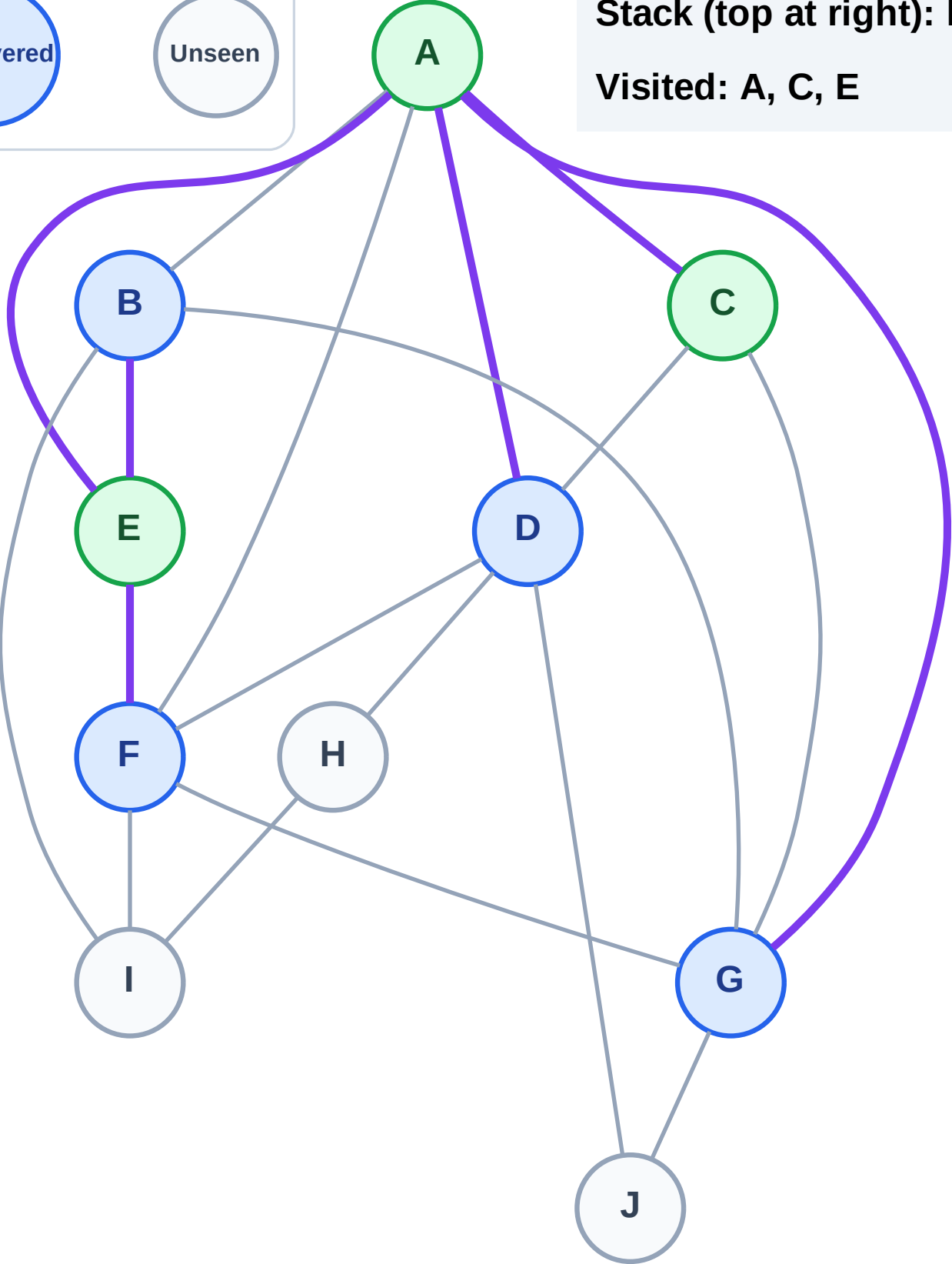
Complete

Processing

Discovered

Unseen

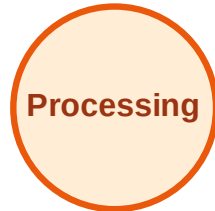
Stack (top at right): F | B | G | D  
Visited: A, C, E



# DFS Traversal

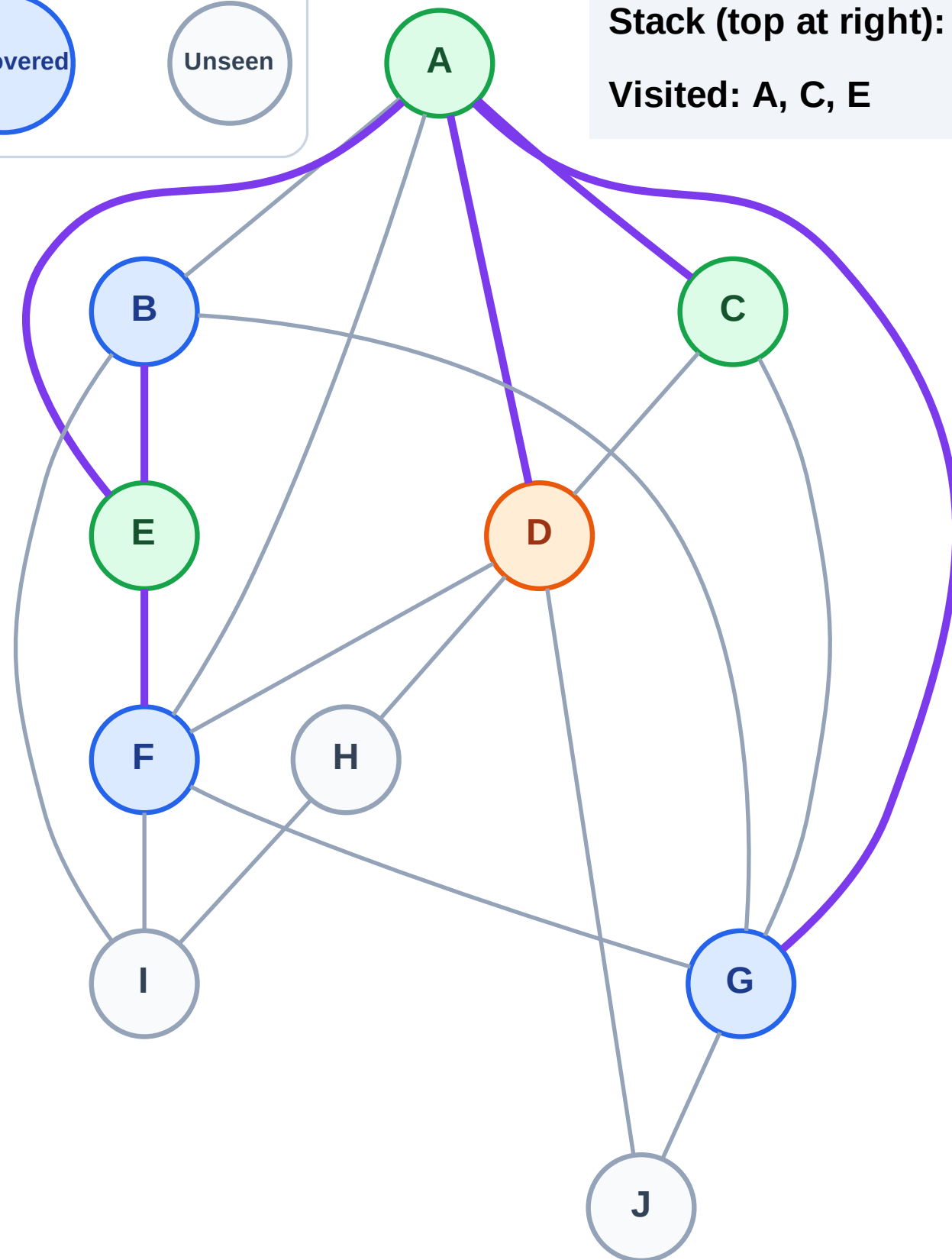
Step 8: Process node D

## Legend



Stack (top at right): F | B | G

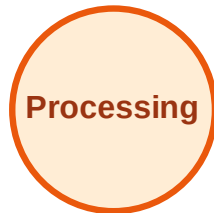
Visited: A, C, E



# DFS Traversal

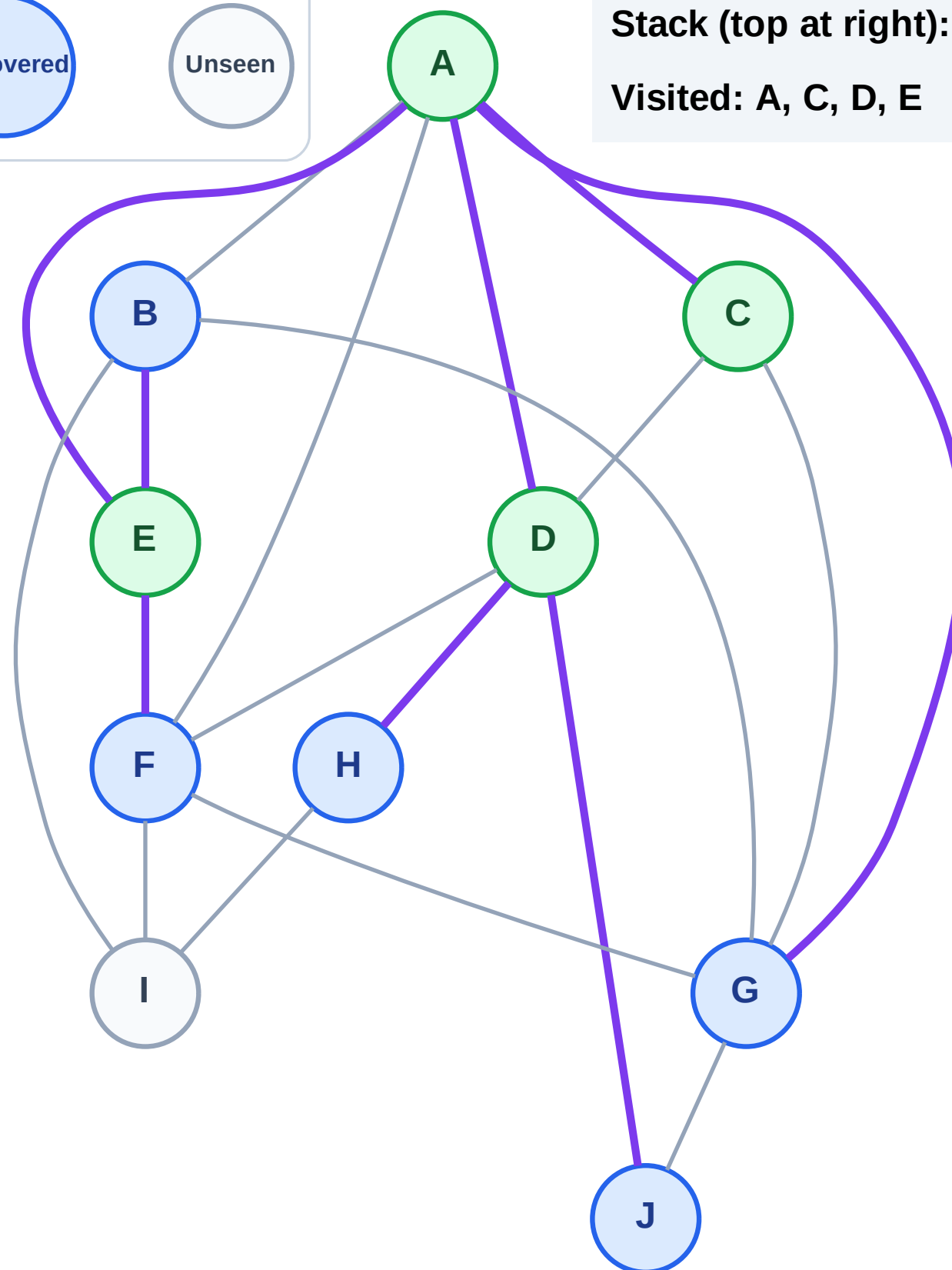
Step 9: Discover J, H from D

## Legend



Stack (top at right): F | B | G | J | H

Visited: A, C, D, E



# DFS Traversal

Step 10: Process node H

Legend

Complete

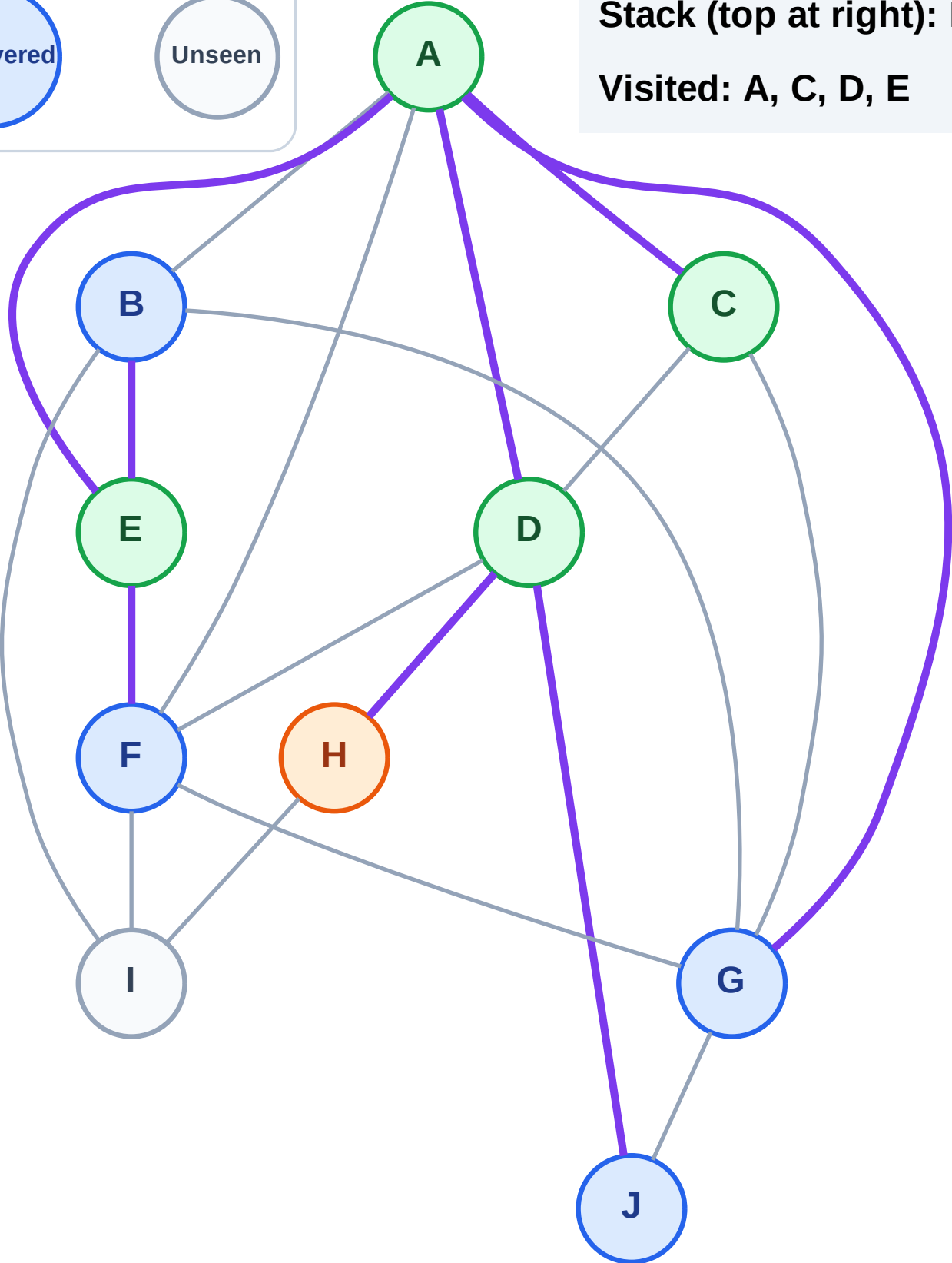
Processing

Discovered

Unseen

Stack (top at right): F | B | G | J

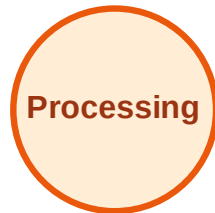
Visited: A, C, D, E



# DFS Traversal

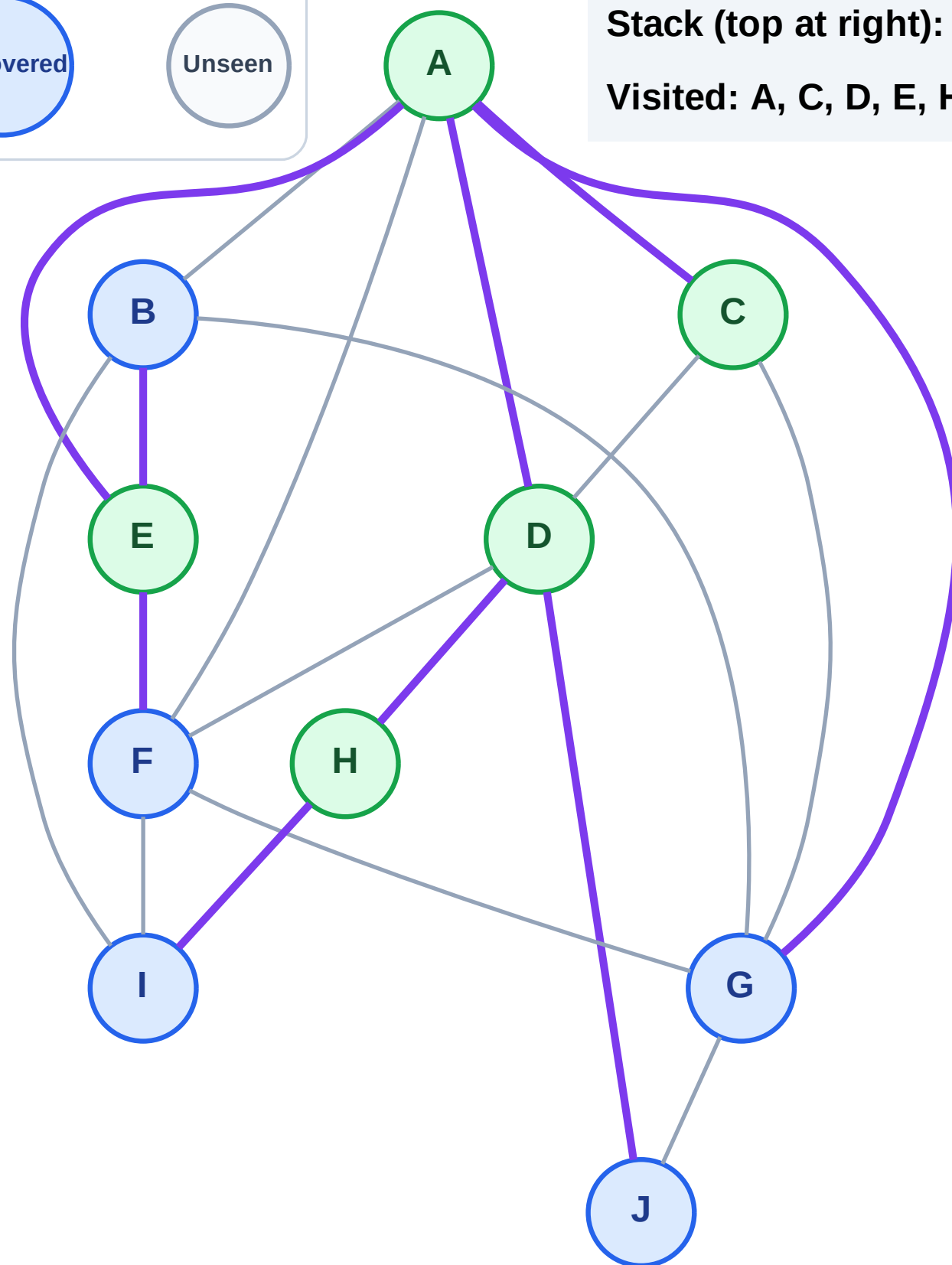
Step 11: Discover I from H

## Legend



Stack (top at right): F | B | G | J | I

Visited: A, C, D, E, H



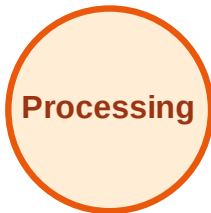
# DFS Traversal

Step 12: Process node I

## Legend



Complete



Processing



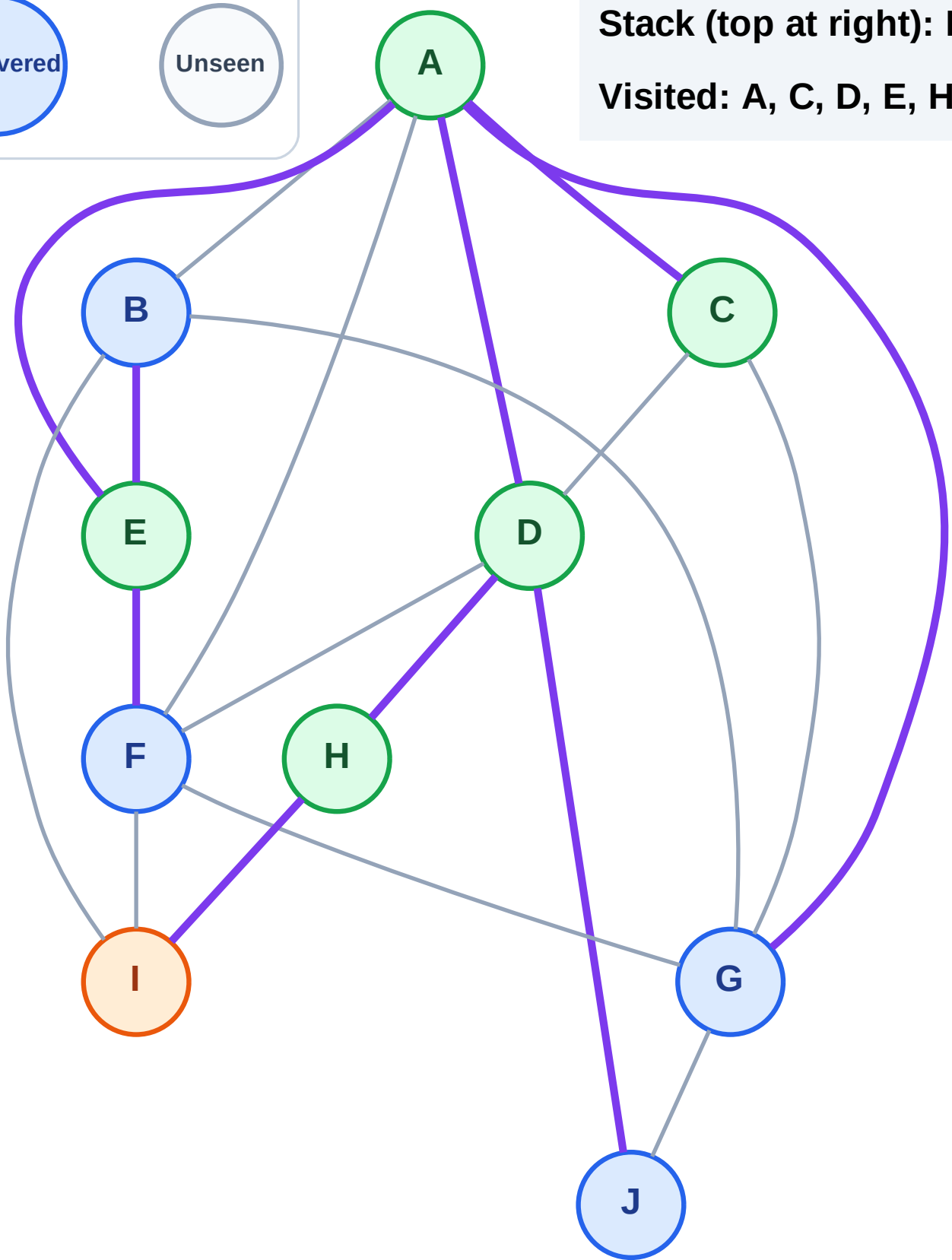
Discovered



Unseen

Stack (top at right): F | B | G | J

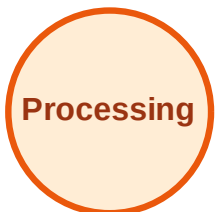
Visited: A, C, D, E, H



# DFS Traversal

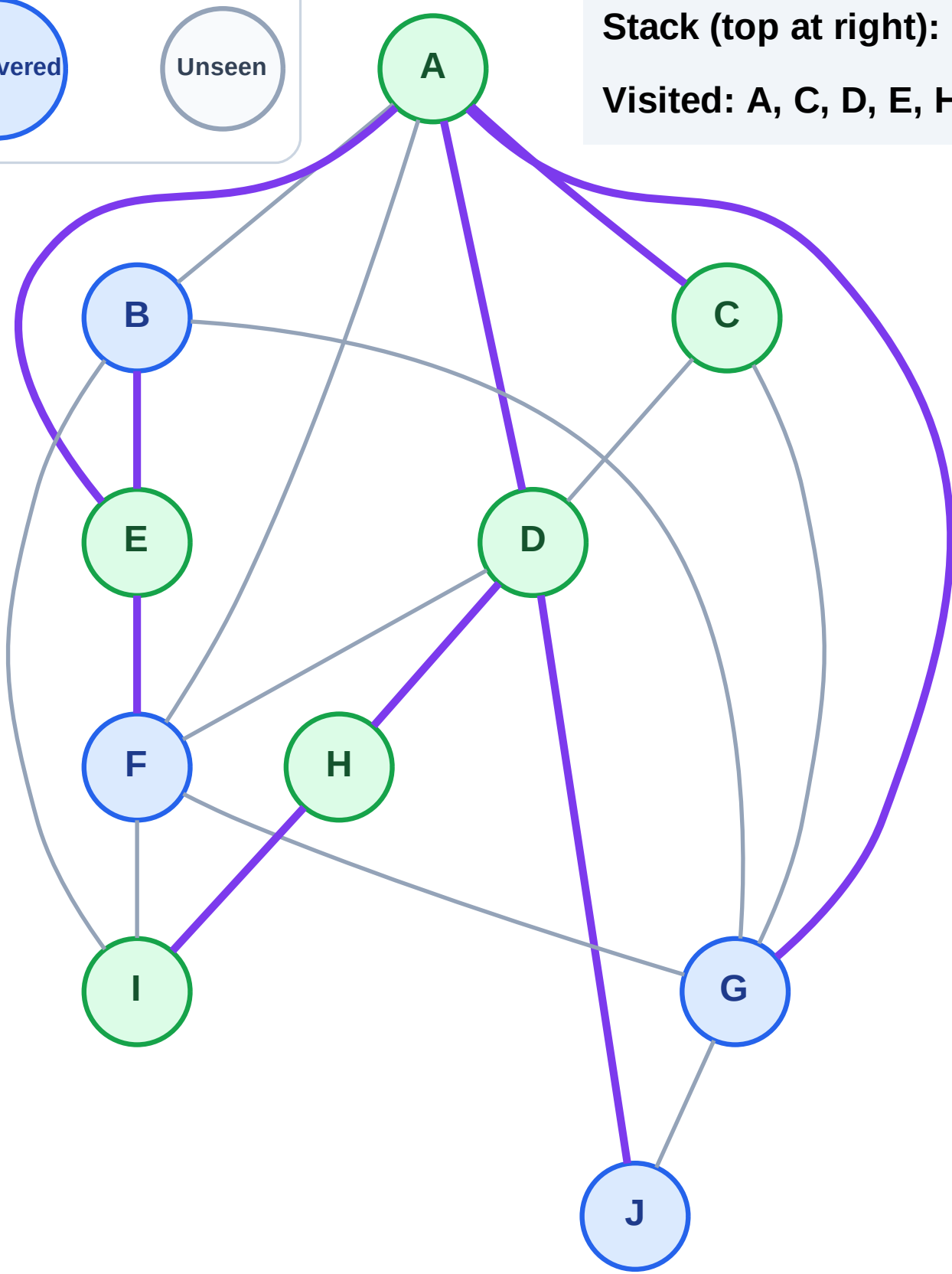
Step 13: No new neighbors from I

## Legend



Stack (top at right): F | B | G | J

Visited: A, C, D, E, H, I



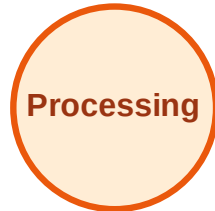
# DFS Traversal

Step 14: Process node J

## Legend



Complete



Processing



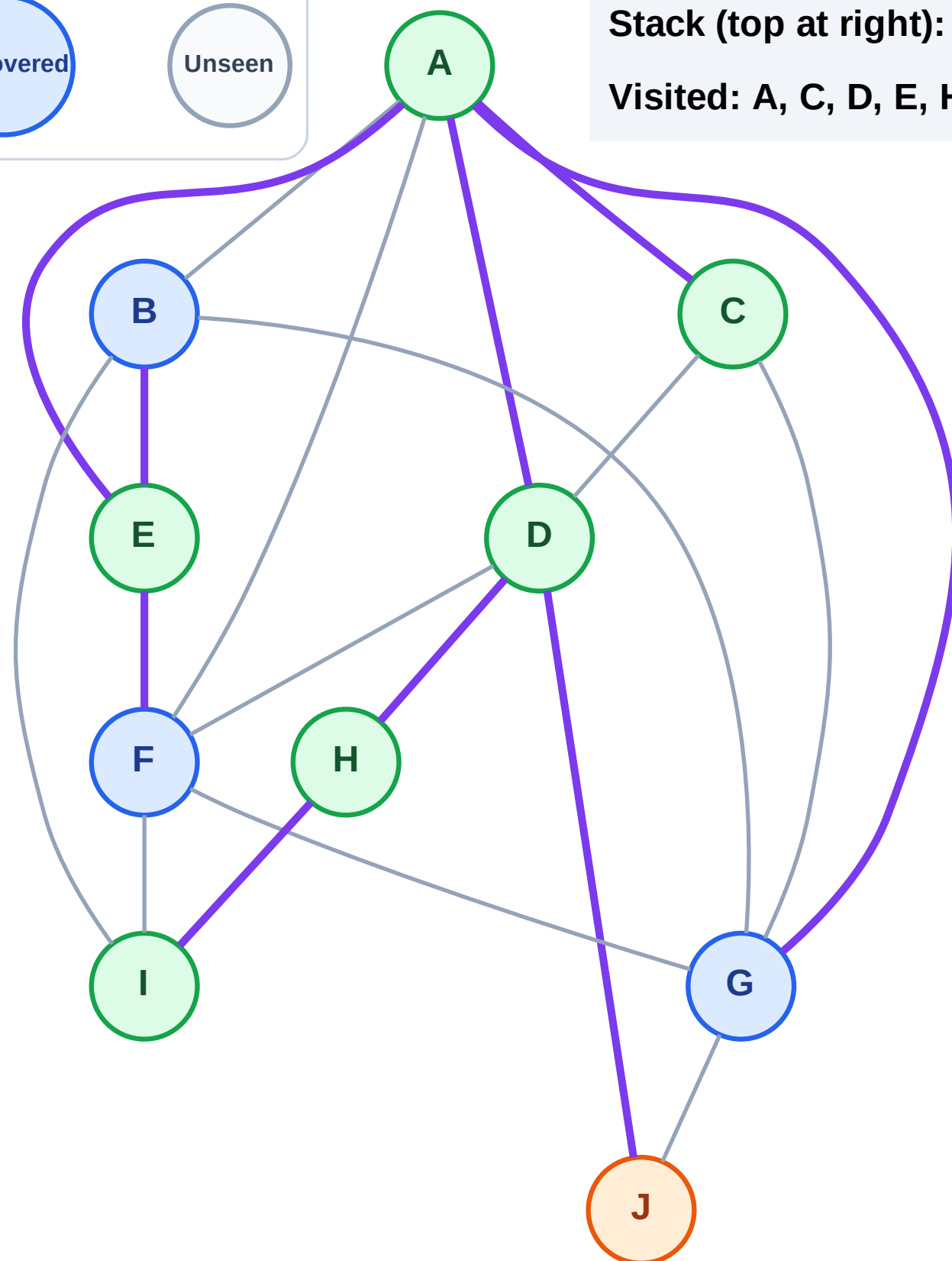
Discovered



Unseen

Stack (top at right): F | B | G

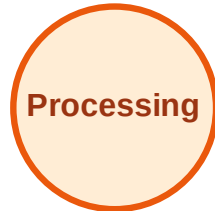
Visited: A, C, D, E, H, I



# DFS Traversal

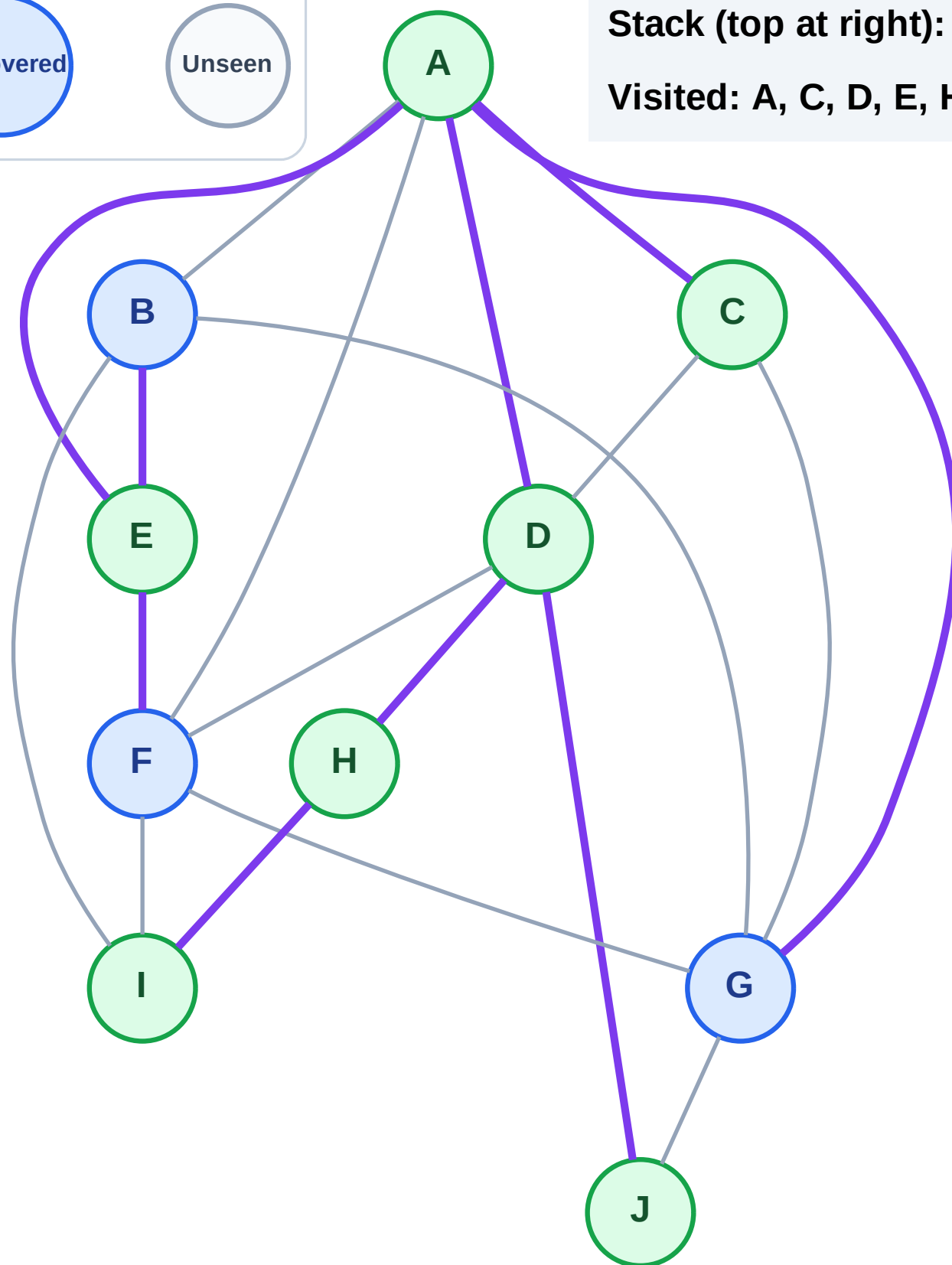
Step 15: No new neighbors from J

## Legend



Stack (top at right): F | B | G

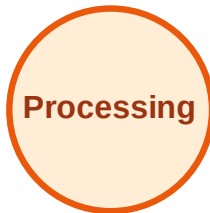
Visited: A, C, D, E, H, I, J



# DFS Traversal

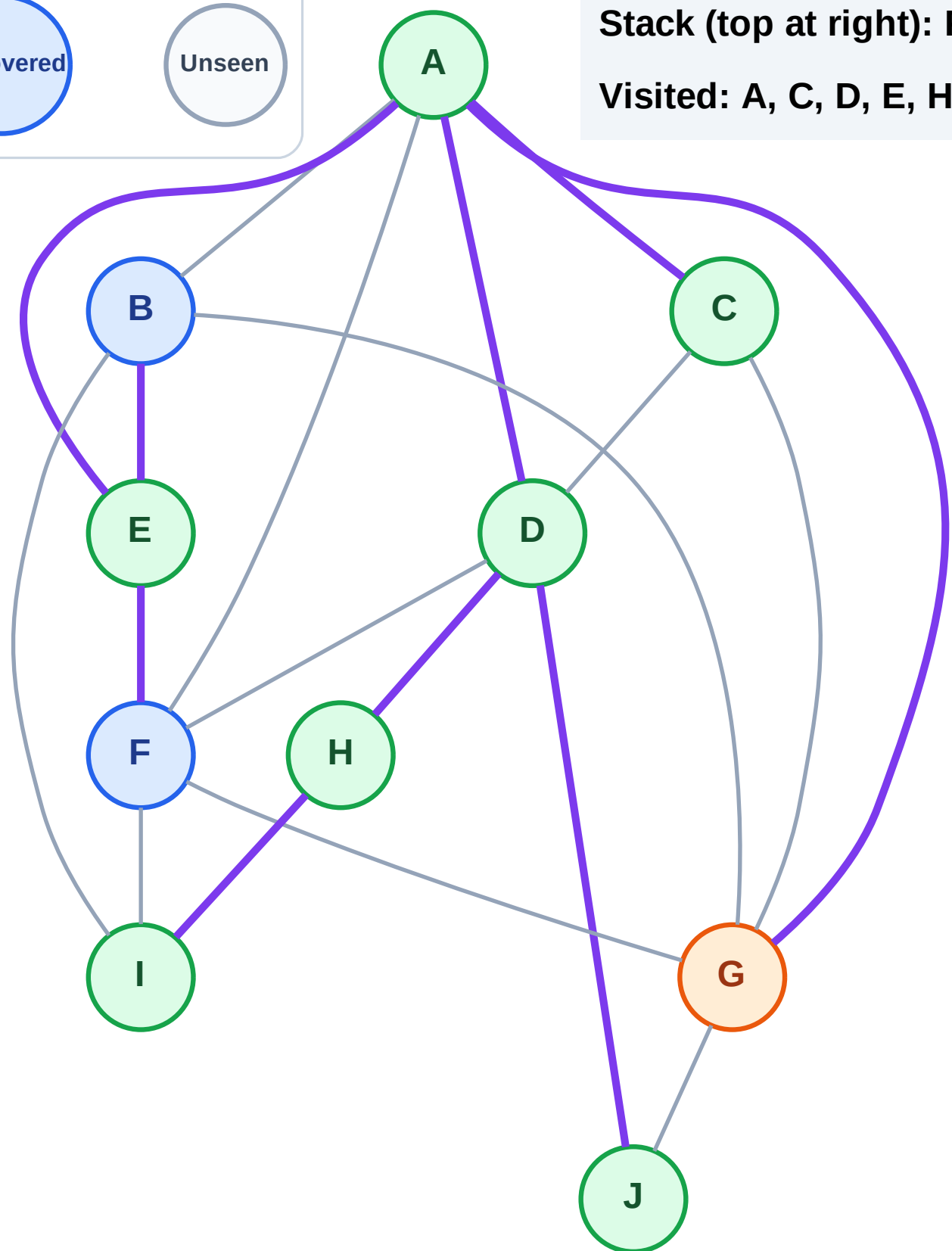
Step 16: Process node G

## Legend



Stack (top at right): F | B

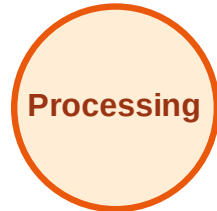
Visited: A, C, D, E, H, I, J



# DFS Traversal

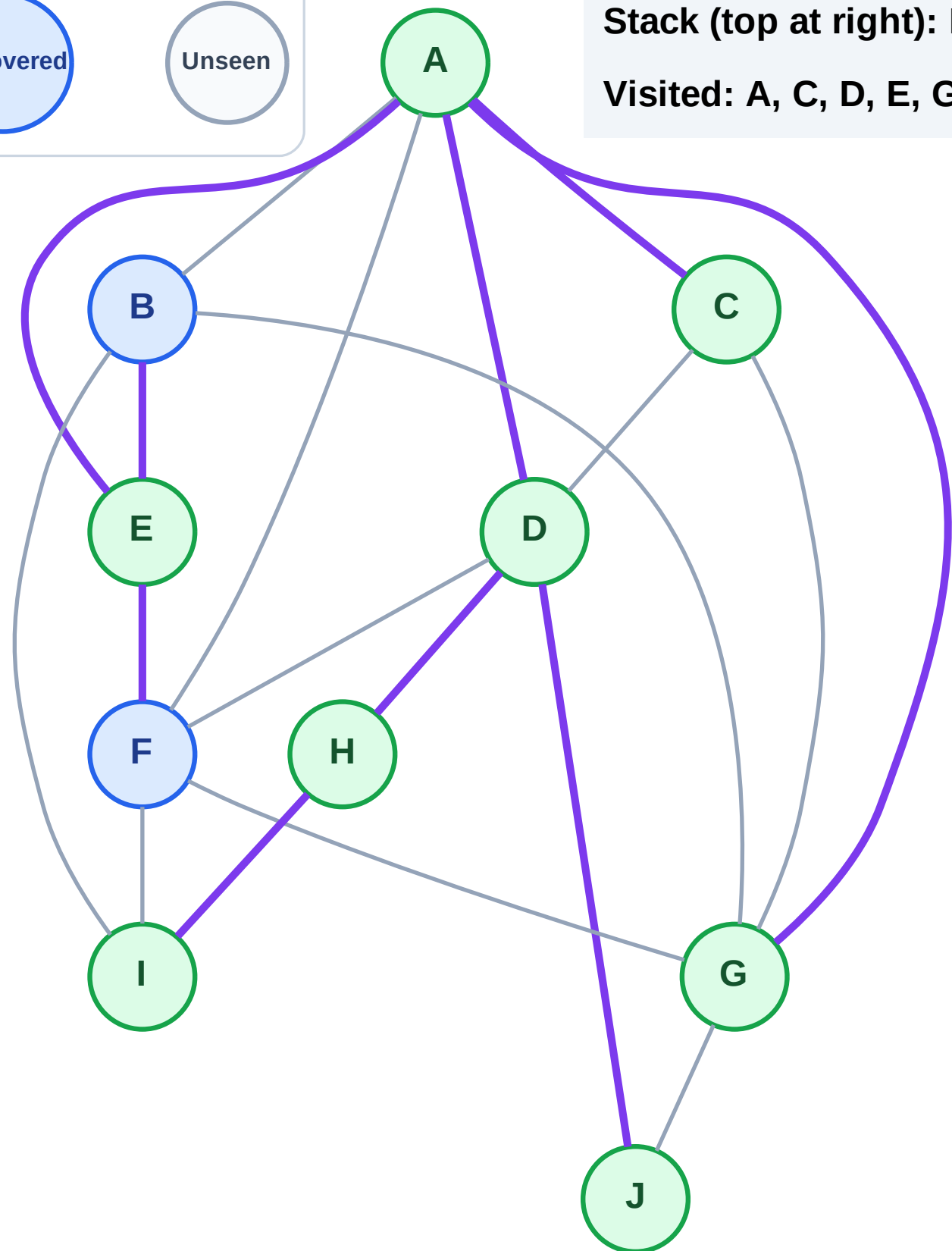
Step 17: No new neighbors from G

## Legend



Stack (top at right): F | B

Visited: A, C, D, E, G, H, I, J



# DFS Traversal

Step 18: Process node B

Legend

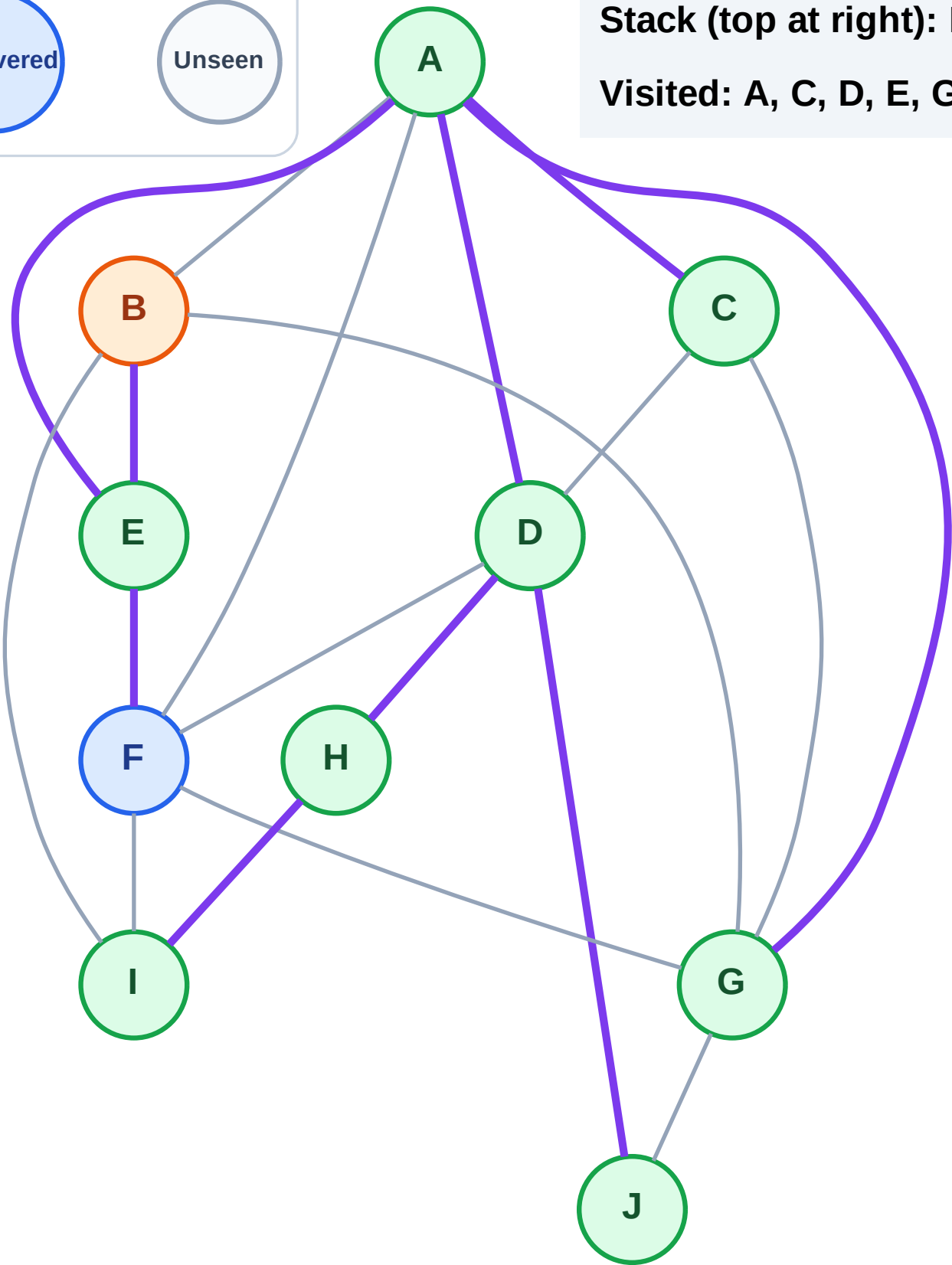
Complete

Processing

Discovered

Unseen

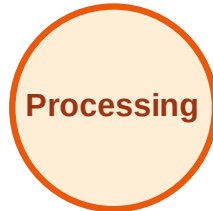
Stack (top at right): F  
Visited: A, C, D, E, G, H, I, J



# DFS Traversal

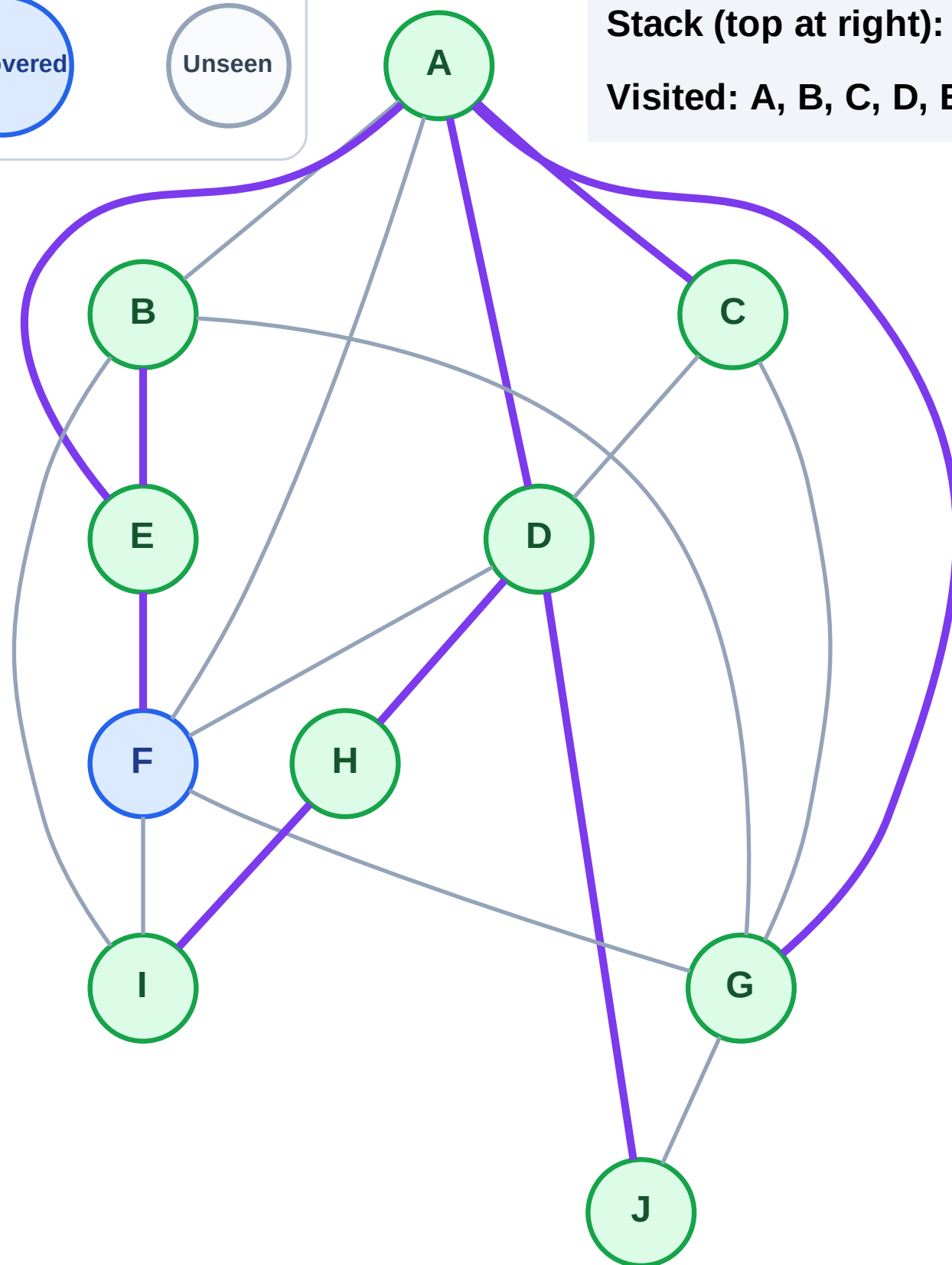
Step 19: No new neighbors from B

## Legend



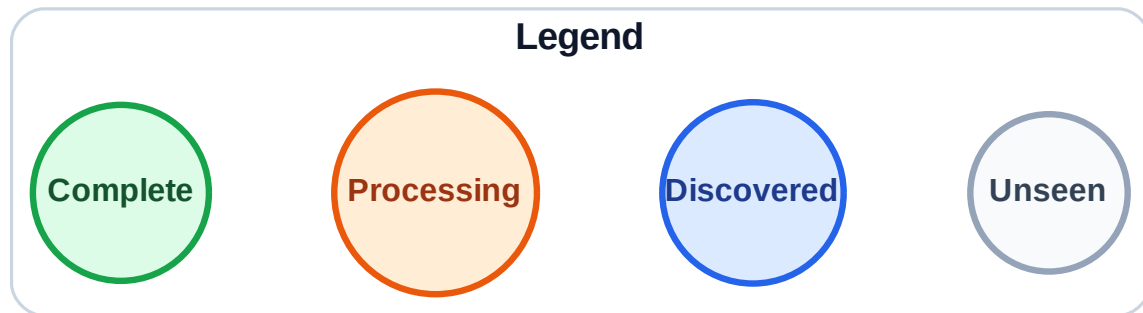
Stack (top at right): F

Visited: A, B, C, D, E, G, H, I, J



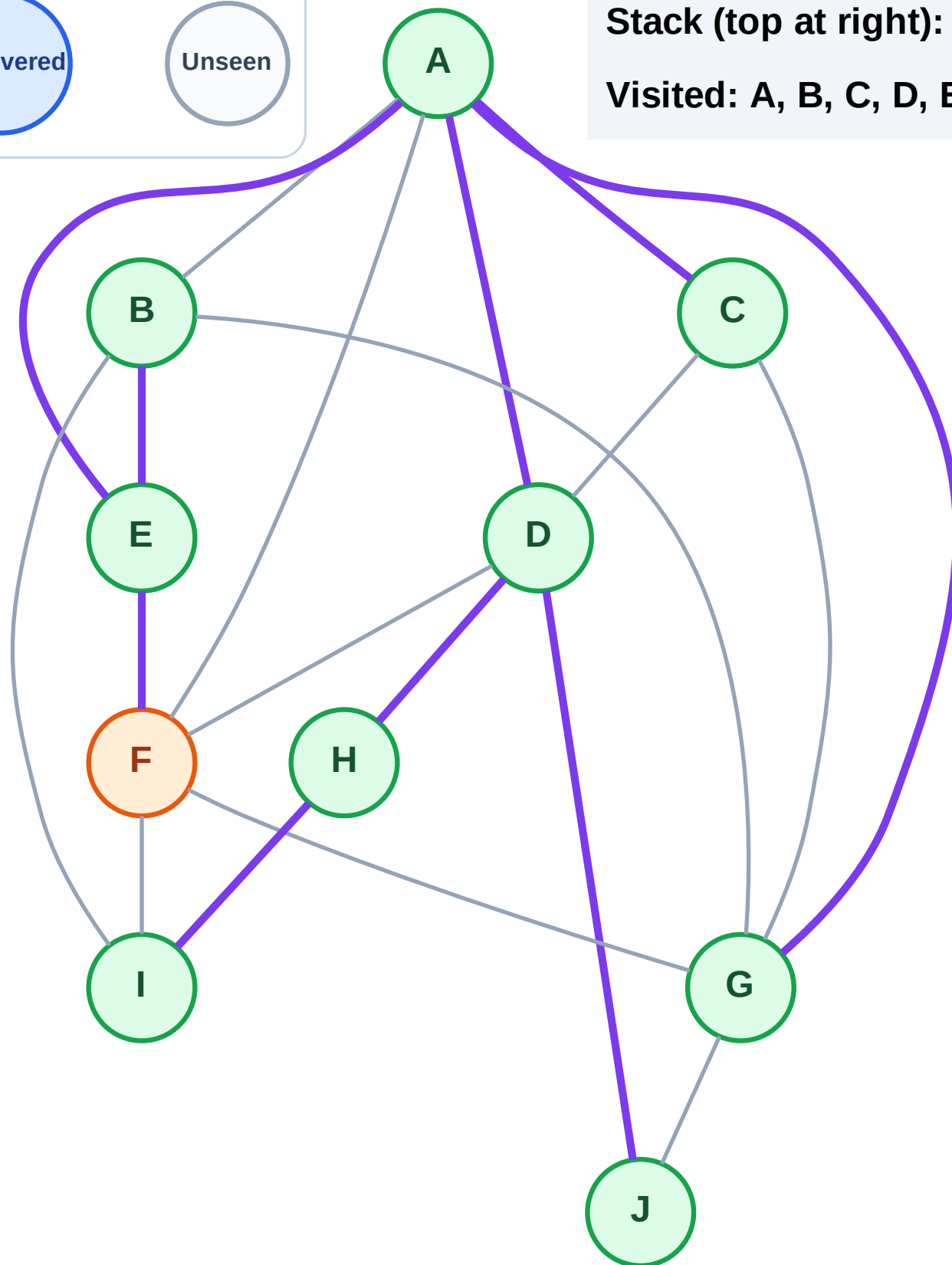
### Step 20: Process node F

### Step 20: Process node F



**Stack (top at right): (empty)**

**Visited: A, B, C, D, E, G, H, I, J**



# DFS Traversal

Step 21: No new neighbors from F

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

